MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

ARITHMETIC LOGIC UNIT, DIGITAL, KEY CODE KDJ

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic arithmetic logic unit.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of the specification and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Space and Naval Warfare Systems Command, ATTN: SPAWAR 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.1 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KDJ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/12-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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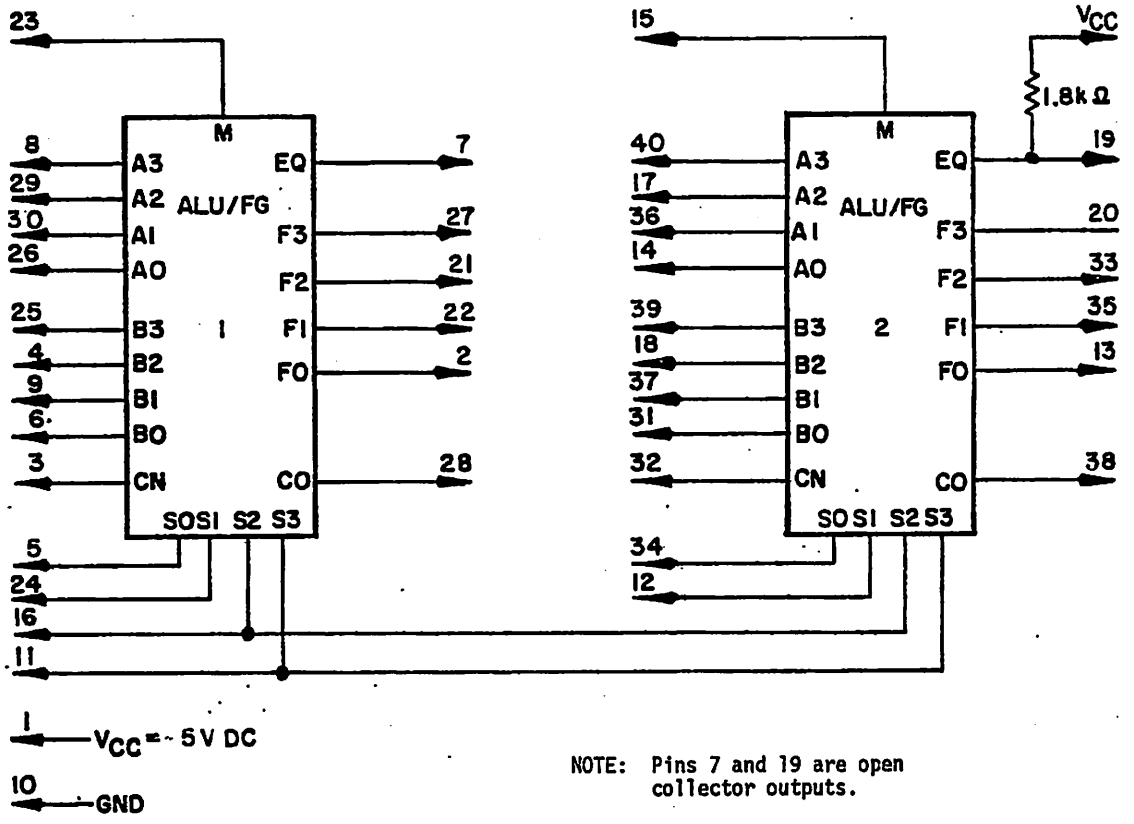


FIGURE 1. KDJ module logic diagram.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	V_{OUT} (low-loaded)	V	S	10	800	10	800	10	800	10	800	mV
B1-B26	Input breakdown current	VI	S	0	1	0	1	0	1	0	1	mA
C	Isolation	VII	S	10	∞	1	∞	10	∞	1	∞	MΩ
D1-D26	Input clamp diode voltage	VIII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E12	Output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F12	Output voltage, low level	X	X	0	400	0	400	0	400	0	400	mV
G1-G2	Input current, low level (M)	XI	X	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	mA
G3-G18	Input current, low level (A or B)	XI	X	1.20	5.17	1.20	5.17	1.20	5.17	1.20	5.17	mA
G19-G22	Input current, low level (S0-S1)	XI	X	1.6	6.9	1.6	6.9	1.6	6.9	1.6	6.9	mA
G23, G24	Input current, low level (S2, S3)	XI	X	3.2	13.8	3.2	13.8	3.2	13.8	3.2	13.8	mA
G25, G26	Input current, low level (CN)	XI	X	2.00	8.62	2.00	8.62	2.00	8.62	2.00	8.62	mA
H1, H2	Input current, high level (M)	XII	X	0	40	0	40	0	40	0	40	µA
H3-H18	Input current, high level (A or B)	XII	X	0	120	0	120	0	120	0	120	µA
H19-H22	Input current, high level (S0,S1)	XII	X	0	160	0	160	0	160	0	160	µA
H23, H24	Input current, high level (S2,S3)	XII	X	0	320	0	320	0	320	0	320	µA
H25, H26	Input current, high level (CN)	XII	X	0	200	0	200	0	200	0	200	µA
J1-J10	Output short circuit current	XIII	X	20	55	20	55	20	55	20	55	mA
J11	Output short circuit current (EQ output)	XIII	X	2.4	3.4	2.4	3.4	2.4	3.4	2.4	3.4	mA
J12	Output short circuit current (EQ output)	XIII	X	0	500	0	500	0	500	0	500	µA
K1	Power supply current, high level	XIV	S	1	254	1	254	1	254	1	254	mA
K2	Power supply current, low level	XIV	S	1	273	1	273	1	273	1	273	mA
L1-L10	Transition time, t_{TLH} (except EQ output)	XV	S	4	45	4	45	4	50	4	50	ns
L11, L12	Transition time, t_{TLH} (EQ output)	XV	S	5	45	5	45	5	50	5	50	ns
L13-L17, L19, L21-L24	Transition time, t_{THL} (except EQ output)	XV	S	4	20	4	20	4	25	4	25	ns
L18, L20	Transition time, t_{THL} (EQ output)	XV	S	1	20	1	20	1	25	1	25	ns
M1, M3, M5, M7, M9 M11, M13, M15	Propagation delay time, t_{PLH} (A to F)	XVI	Q	6	41	6	41	6	43	6	43	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M2, M4, M6, M8, M10, M12, M14, M16	Propagation delay time, t_{PHL} (A to F)	XVI	Q	6	50	6	50	6	52	6	52	ns
M17, M19, M21, M23, M25, M27, M29, M31	Propagation delay time, t_{PLH} (B to F)	XVI	S	6	50	6	50	6	52	6	52	ns
M18, M20, M22, M24, M26, M28, M30, M32	Propagation delay time, t_{PHL} (B to F)	XVI	S	6	41	6	41	6	43	6	43	ns
M33-M40	Propagation delay time, t_{PLH} (CN to F)	XVI	S	4	27	4	27	4	29	4	29	ns
M41, M42	Propagation delay time, t_{PLH} (CN to CO)	XVI	S	4	26	4	26	4	28	4	28	ns
M43, M44	Propagation delay time, t_{PHL} (CN to CO)	XVI	S	4	28	4	28	4	30	4	30	ns
M45-M52	Propagation delay time, t_{PHL} (CN to F)	XVI	S	4	27	4	27	4	29	4	29	ns
M53-M68	Propagation delay time, t_{PLH}, t_{PHL} (A to EQ)	XVI	Q	6	63	6	63	6	65	6	65	ns
M69-M84	Propagation delay time, t_{PLH}, t_{PHL} (B to EQ)	XVI	S	6	63	6	63	6	65	6	65	ns
N1, N24	Input capacitance	XVII	Q	2	18	2	18	2	18	2	18	pF
N25,N26	Input capacitance (S2,S3)	XVII	Q	2	35	2	35	2	35	2	35	pF
P	Power supply transient rise time	XVIII	S	1	2	1	2	1	2	1	2	ns
R1-R4, R6-R9, R15, R22-R24	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	mV
R5, R10	Output crosstalk, low level	XIX	Q	0	600	0	600	0	600	0	600	mV
R11-R14, R16-R21	Output crosstalk, low level	XIX	Q	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XX	Q	0	400	0	400	0	400	0	400	mV
T	Functional test	-	X	See paragraph 4.6								

1/ "X" parameters shall be 100 percent tested on all modules. Modules which have not previously been used in an operating system, or have been in storage for less than 2 years, shall meet the 25°C ±5°C initial acceptance requirements of table I.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater at 25°C).
3/ "Q" parameters shall be tested during qualification testing only.

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* 4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, B, C, D, and J are stress tests and will be performed only at 25°C ±5°C. If test conditions A, B, C, D, or J are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, J where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through H, K through M, P, and T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition P, shall be performed using the initial limits at 25°C ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test, and while still at the maximum class temperature, electrical tests E through H and K through T shall be performed in the order specificed in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C ±5°C, electrical tests E through H and K through T shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

4.6 Functional tests. All modules shall be subjected to functional tests which conform to the requirements stated in 4.6.1.

4.6.1 Test patterns. A set of input patterns shall be generated capable of detecting all nonredundant failures in the module logic network. The method of generating the required test patterns is as follows:

a. The module logic network shall be represented in terms of three basic logic functions - AND, OR, INVERT.

b. The test patterns shall be generated by assuming certain failures in each of the basic logic functions and applying input patterns to detect those failures as shown in table II. The test patterns necessary to satisfy these requirements and the logic representation shall be submitted with the design qualification samples.

4.6.2 Input requirements. The following input voltage levels shall be used for the test patterns:

a. The low-level input voltage shall be 800 mV ±1 percent.

b. The high-level input voltage shall be 2 V ±1 percent.

4.6.3 Output requirements. The output requirements for the test patterns shall be as follows:

a. The low-level output voltage shall be 0 V minimum and 400 mV maximum.

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b. The high-level output voltage shall be 2.4 V minimum and 4.5 V maximum for all pins except 7 and 19.

c. The high-level output voltage shall be 2.4 V minimum and 5.5 V maximum for pins 7 and 19.

4.6.4 Test configuration. The test equipment, power supply, and loading requirements shall be as specified on figure 9.

TABLE II. Test patterns. 1/, 2/

Function	Failure	Input pattern						Fault free output
		X1	X2	X3	...	XN-1	XN	Z
X1, X2, ., XN — AND — Z	X1 Stuck 1	0	1	1	...	1	1	0
	X2 Stuck 1	1	0	1	...	1	1	0
	.	.	.	.	.	.	.	.
	.	.	.	.	.	.	.	.
	.	.	.	.	.	.	.	.
	XN Stuck 1	1	1	1		1	0	0
	Z Stuck 0	1	1	1	...	1	1	1
X1, X2, ., XN — OR — Z	X1 Stuck 0	1	0	0	...	0	0	1
	X2 Stuck 0	0	1	0	...	0	0	1
	.	.	.	.	.	.	.	.
	.	.	.	.	.	.	.	.
	.	.	.	.	.	.	.	.
	XN Stuck 0	0	0	0	...	0	1	1
	Z Stuck 1	0	0	0	...	0	0	0
X1 — INV — Z	Z Stuck 1	1	-	-	-	-	-	0
	Z Stuck 0	0	-	-	-	-	-	1

1/ Stuck 1 indicates a failure resulting in a high level.
2/ Stuck 0 indicates a failure resulting in a low level.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table III, the module is considered to have completed its useful life.

6.3 Cross reference of classification. Military Specification MIL-M-28787/12 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833409. Part number M28787/12-1 supersedes part numbers 2833409 and M28787/12.

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* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits of table I.

Supply voltage	5 V dc ±10%
Maximum power dissipation	1.5 W (outputs not loaded)
Input voltage	
Low level	800 mV maximum
High level	2 V maximum
Mode control input current	
Low level (V_{IL} = 0 V)	-1.73 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
A or B Data input current	
Low level (V_{IL} = 0 V)	-5.17 mA maximum
High level (V_{IH} = 2.4 V)	120 µA maximum
Independent select line input current	
Low level (V_{IL} = 0 V)	-6.9 mA maximum
High level (V_{IH} = 2.4 V)	160 µA maximum
Common select line input current	
Low level (V_{IL} = 0 V)	-13.8 mA maximum
High level (V_{IH} = 2.4 V)	320 µA maximum
Carry input current	
Low level (V_{IL} = 0 V)	-8.62 mA maximum
High level (V_{IH} = 2.4 V)	200 µA maximum
Output voltage	
Low level	400 mV maximum
High level	
All F and CO	2.4 V minimum, V_{CC} maximum
All EQ	2.4 V minimum, V_{CC} maximum
Output drive	
Low level (V_{OL} = 400 mV)	
All outputs except pin 19 and 7	16 mA maximum
Pin 19 and 7	12.8 mA maximum
High level (V_{OH} = 2.4 V)	
All F and CO	800 µA minimum
EQ (Pin 19)	800 µA minimum
EQ (Pin 7)	Open collector output

* 6.5 Module function. This module contains two ALU/FG's.

6.5.1 Input and output relationship. Each ALU/FG will perform either logic functions (M = "1") or arithmetic functions (M ="0"). The function performed on the A and B inputs is determined by the logic state of the select inputs (S0, S1, S2, and S3). Logic functions are performed on a bit basis. All arithmetic functions are performed on a word basis where each word consists of four bits. The A word is composed of bits A0, A1, A2, and A3, where A0, is the least significant bit. The B word is composed of bits B0, B1, B2, and B3, where B0 is considered the least significant bit. The carry input (CN) and the ripple carry output (CO) are provided for arithmetic operations. Placing the ALU/FG in the subtract mode (S0 = 0, S1 = 1, S2 = 1, and S3 = 0) produces a CO output that represents A and B word magnitude comparisons. The following table III exists for the magnitude comparison.

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TABLE III. Input and output relationships.

Word comparisons	Input	Output
	CN	CO
$A \leq B$	1	1
$A < B$	0	1
$A > B$	1	0
$A \geq B$	0	0

When two words of equal magnitude are applied to the A and B inputs, the EQ output will go to a logic "1". For all other cases, the EQ output will be a logic "0". When the ALU/FG is operated in the Logic mode (M = 1), the CO output is considered a don't care. Each ALU/FG shall have the following input and output relationship of table IV.

TABLE IV. ALU/FG input and output relationship. 1/ - 5/

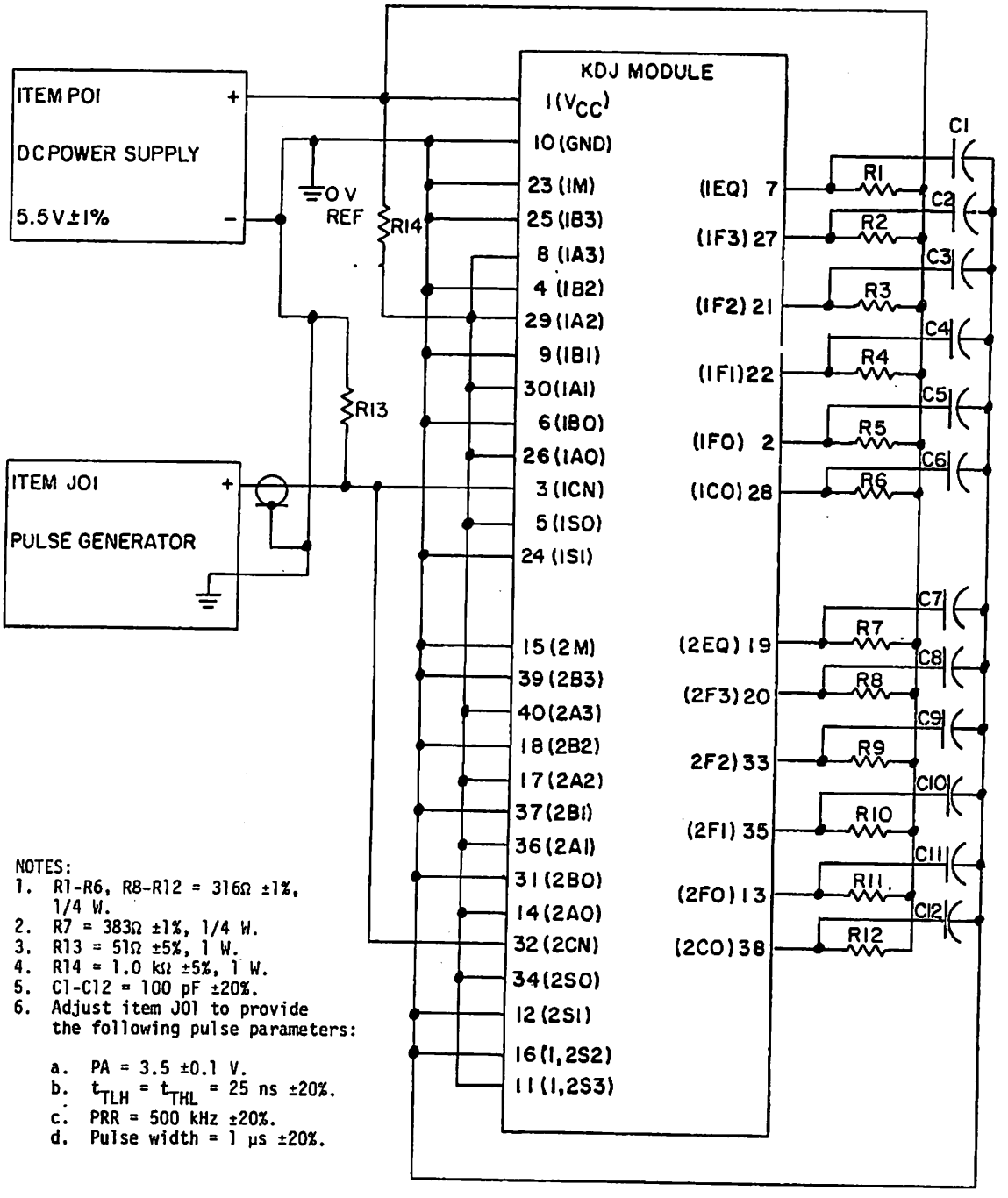
Selection				M = 1 Logic functions	M = 0, arithmetic operations	
S3	S2	S1	S0		CN = 1	CN = 0
0	0	0	0	$F = \overline{A}$	F = A	F = A PLUS 1
0	0	0	1	$F = \overline{A + B}$	F = A + B	F = (A + B) PLUS 1
0	0	1	0	$F = \overline{A} \cdot B$	F = A + $\overline{B}$	F = (A + $\overline{B}$) PLUS 1
0	0	1	1	F = 0	F = MINUS 1 (two's comp.)	F = ZERO
0	1	0	0	$F = \overline{A \cdot B}$	F = A PLUS A·$\overline{B}$	F = A PLUS A·$\overline{B}$ PLUS 1
0	1	0	1	$F = \overline{B}$	F = (A + B) PLUS A·$\overline{B}$	F = (A + B) PLUS A·$\overline{B}$ PLUS 1
0	1	1	0	$F = A \oplus B$	F = A MINUS B MINUS 1	F = A MINUS B
0	1	1	1	$F = A \cdot \overline{B}$	F = A·$\overline{B}$ MINUS 1	F = A·$\overline{B}$
1	0	0	0	$F = \overline{A} + B$	F = A PLUS A·B	F = A PLUS A·B PLUS 1
1	0	0	1	$F = \overline{A \oplus B}$	F = A PLUS B	F = A PLUS B PLUS 1
1	0	1	0	F = B	F = (A + $\overline{B}$) PLUS A·B	F = (A + $\overline{B}$) PLUS A·B PLUS 1
1	0	1	1	F = A·B	F = A·B MINUS 1	F = A·B
1	1	0	0	F = 1	F = A PLUS A 1/	F = A PLUS A PLUS 1
1	1	0	1	$F = A + \overline{B}$	F = (A + B) PLUS A	F = (A + B) PLUS A PLUS 1
1	1	1	0	F = A + B	F = (A + $\overline{B}$) PLUS A	F = (A + $\overline{B}$) PLUS A PLUS 1
1	1	1	1	F = A	F = A MINUS 1	F = A

1/ Each bit is shifted to the next more significant position.
2/ ⊕ Exclusive OR operator
3/ + OR Operator
4/ - Negative operator
5/ . AND operator

6.5.2 Parallel outputs. Paralleling of outputs is not permissible, except for the EQ outputs. To allow comparison of more than two words, paralleling of EQ outputs is permitted providing the following constraints are observed:

EQ output pin 7 is an open collector while EQ output pin 19 is connected to a 1.8 kΩ resistor tied to the output and returned to V_{CC}. When using only EQ output from pin 7, a pullup resistor equal to any value from 316 Ω to 1.82 kΩ must be connected from pin 7 to V_{CC}. When paralleling outputs, a maximum of three pin 7 outputs may be tied to each pin 19 output. No more than four of any EQ outputs may be tied together.

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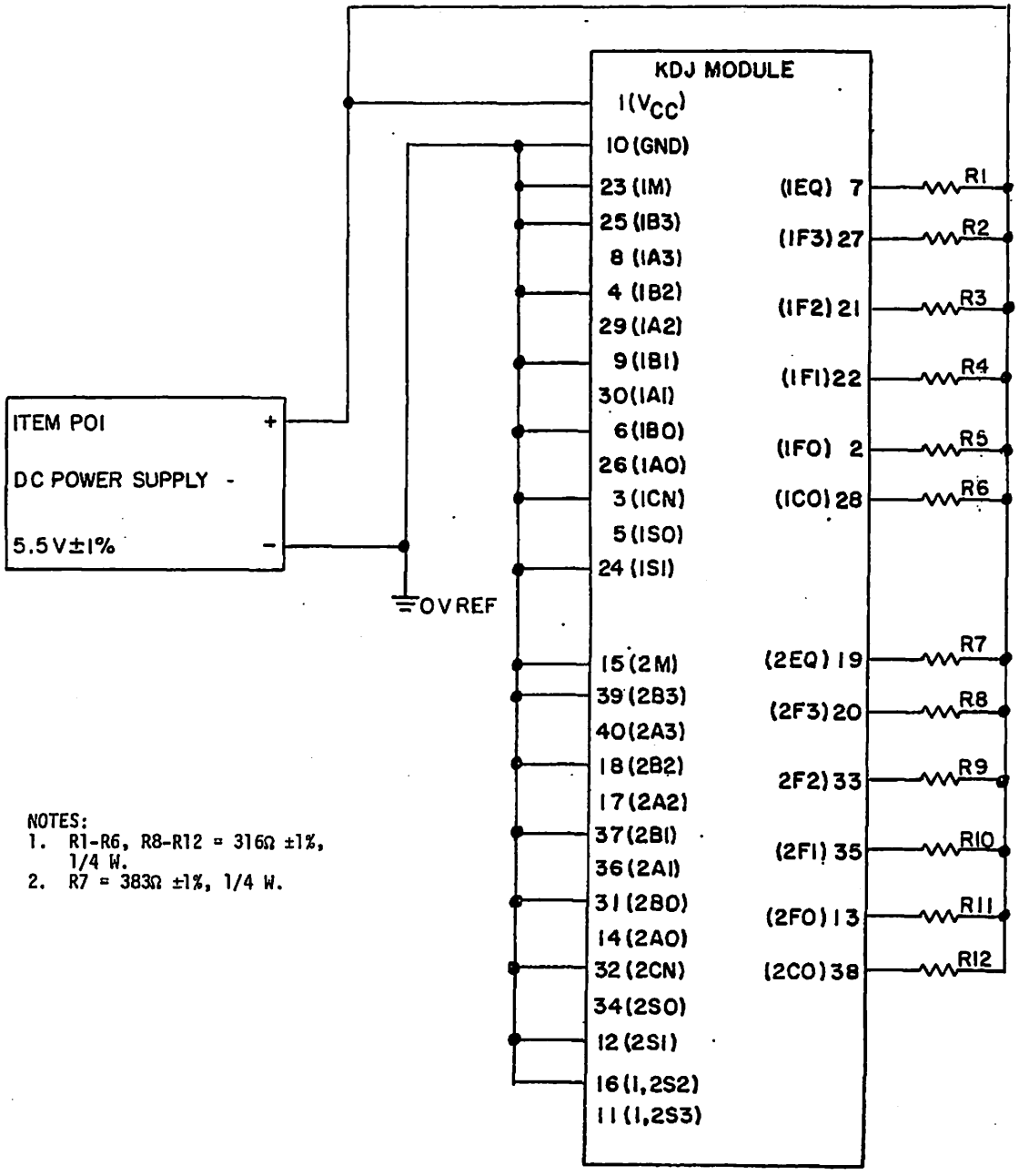


NOTES:
1. R1-R6, R8-R12 = 316Ω ±1%, 1/4 W.
2. R7 = 383Ω ±1%, 1/4 W.
3. R13 = 51Ω ±5%, 1 W.
4. R14 = 1.0 kΩ ±5%, 1 W.
5. C1-C12 = 100 pF ±20%.
6. Adjust item J01 to provide the following pulse parameters:

   a. PA = 3.5 ±0.1 V.
   b. $t_{TLH} = t_{THL}$ = 25 ns ±20%.
   c. PRR = 500 kHz ±20%.
   d. Pulse width = 1 µs ±20%.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1-R6, R8-R12 = 316Ω ±1%, 1/4 W.
2. R7 = 383Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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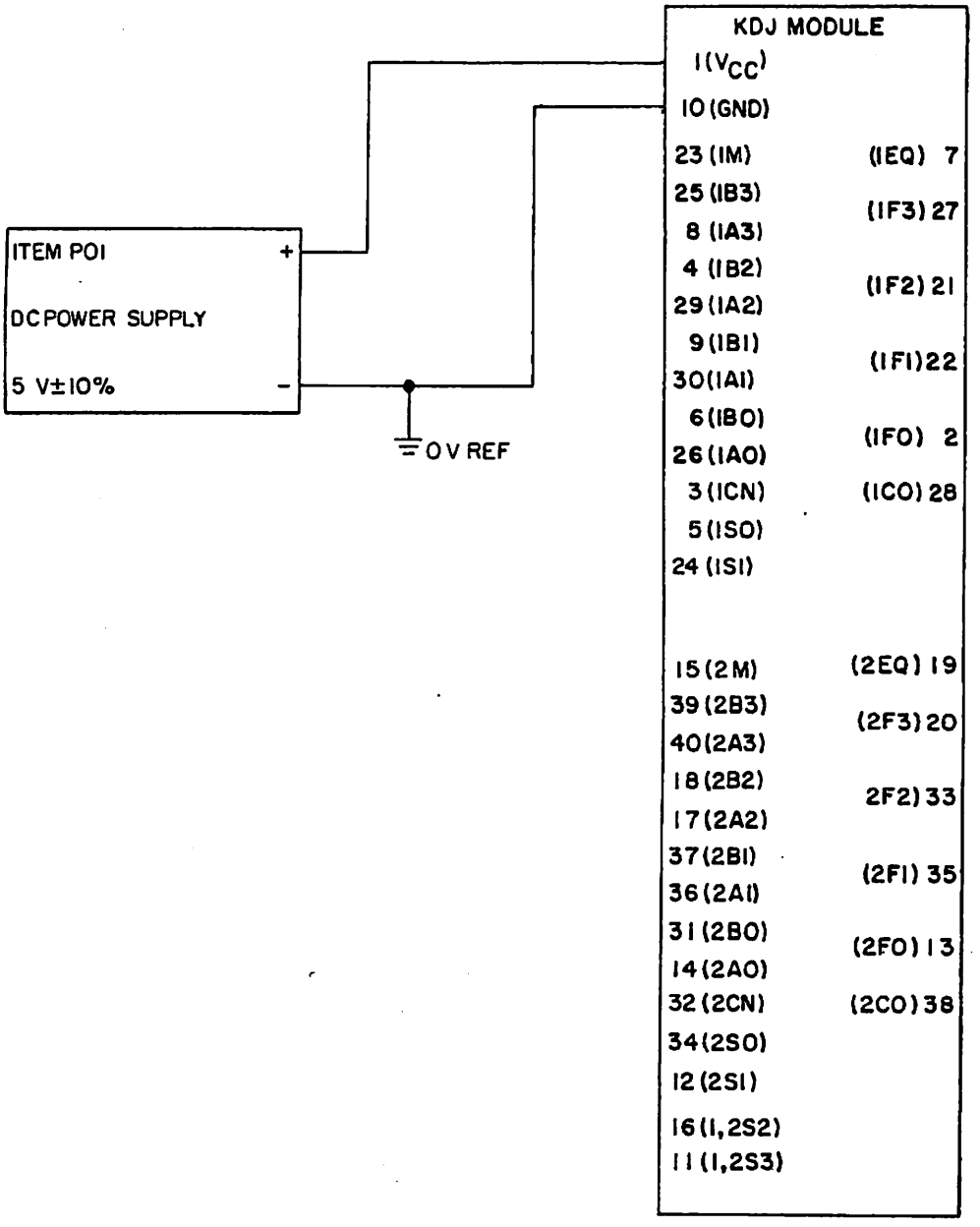


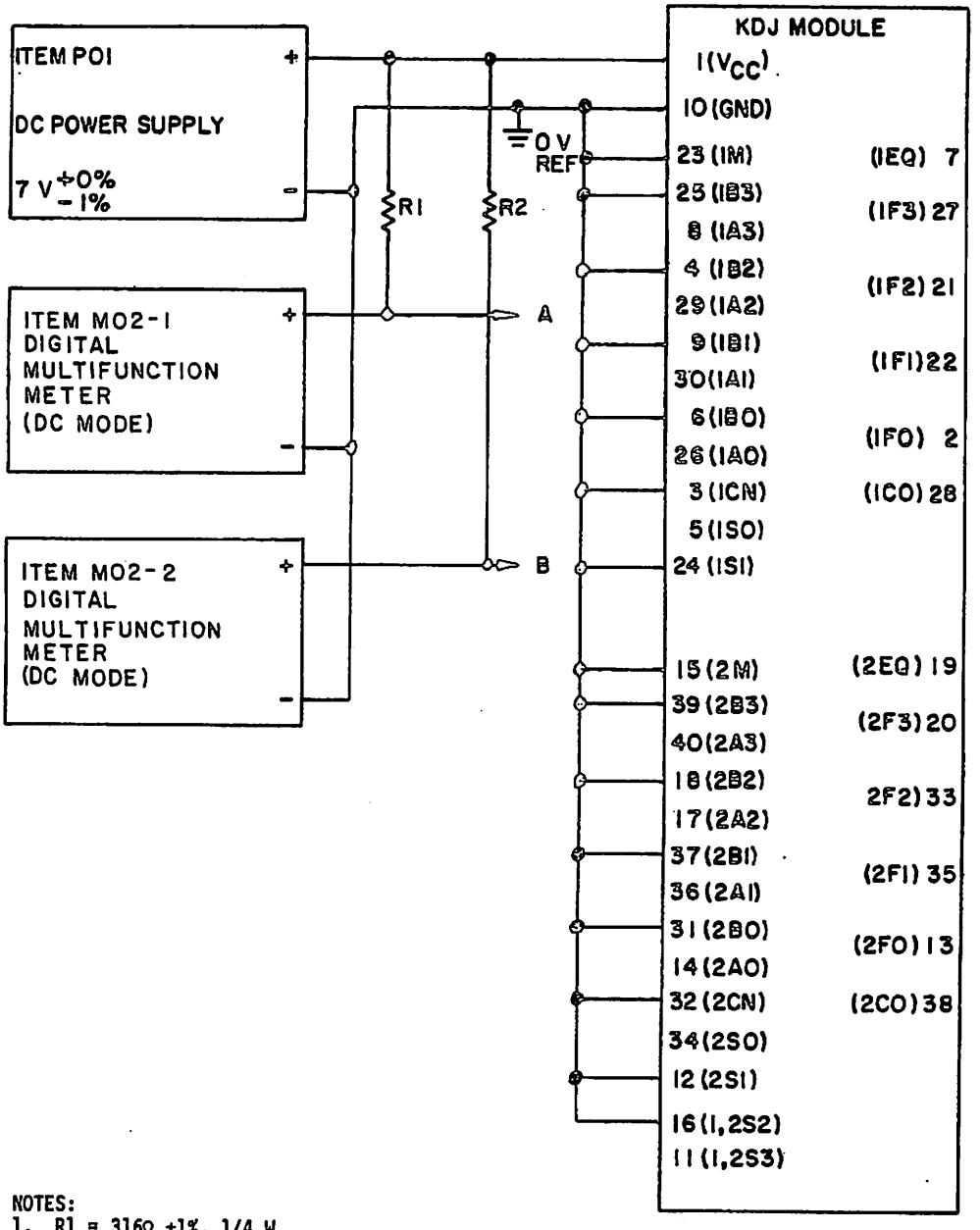
FIGURE 4. Vibration test circuit.

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TABLE V. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	F3 V_{OUT} (low loaded)	27	Open	10	800	mV	5
A2	F2 "	21	"	"	"	"	"
A3	F1 "	22	"	"	"	"	"
A4	F0 "	2	"	"	"	"	"
A5	C0 "	28	"	"	"	"	"
A6	EQ "	7	"	"	"	"	"
A7	C0 "	38	"	"	"	"	"
A8	F3 "	20	"	"	"	"	"
A9	F2 "	33	"	"	"	"	"
A10	F1 "	35	"	"	"	"	"
A11	F0 "	13	"	"	"	"	"
A12	EQ "	Open	19	"	"	"	"

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NOTES:
1. R1 = 316Ω ±1%, 1/4 W.
2. R2 = 383Ω ±1%, 1/4 W.
3. Since the outputs are measured one at a time, only one item M02 is required.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table V.

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TABLE VI. Input breakdown current test.

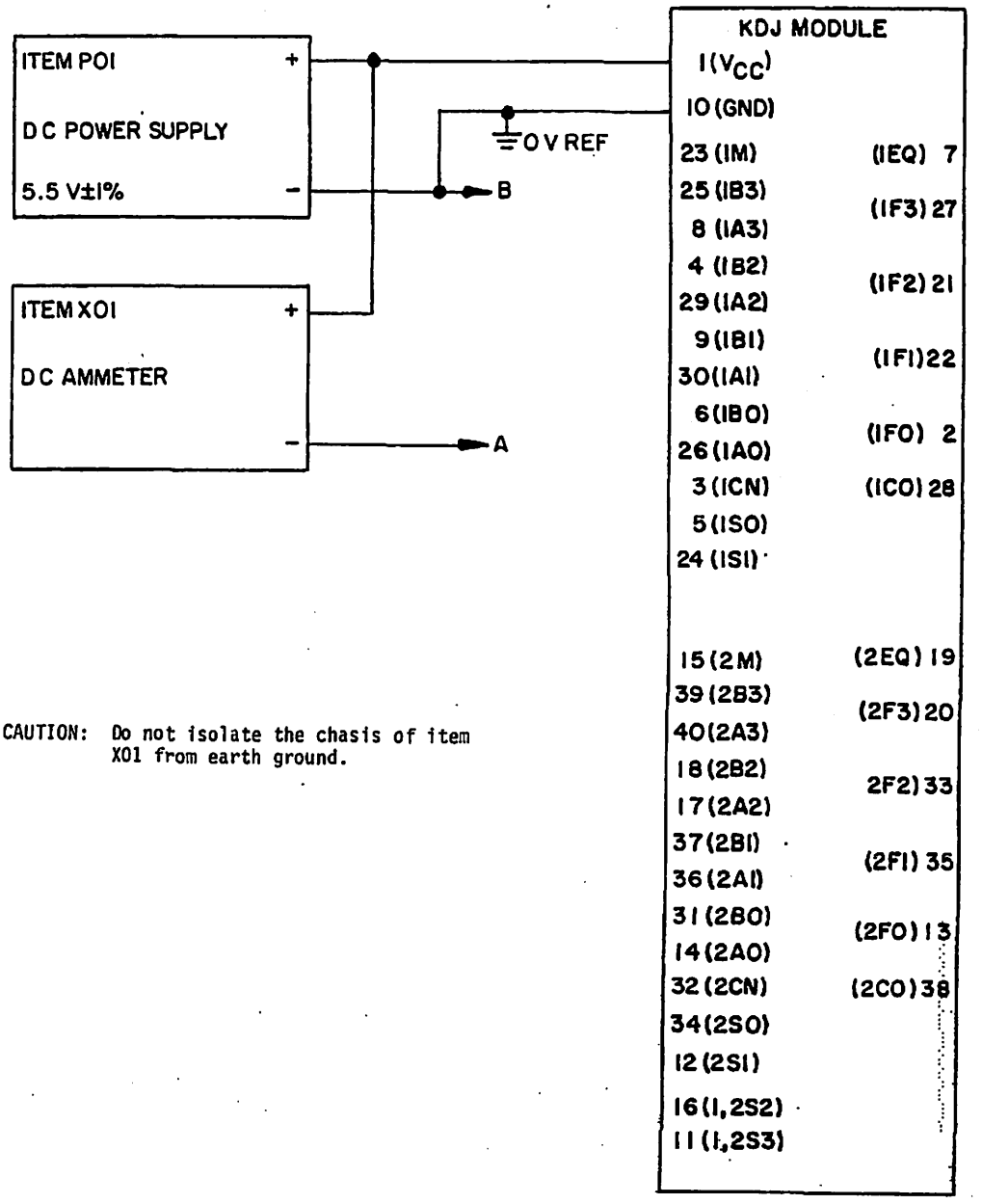
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	B3 Input breakdown current	25	5, 11	0	1	mA	6
B2	A3 "	8	11, 16, 25	"	"	"	"
B3	B2 "	4	5, 11	"	"	"	"
B4	A2 "	29	11, 16, 4	"	"	"	"
B5	B1 "	9	5, 11	"	"	"	"
B6	A1 "	30	11, 16, 9	"	"	"	"
B7	B0 "	6	5, 11	"	"	"	"
B8	A0 "	26	11, 16, 6	"	"	"	"
B9	CN "	3	Open	"	"	"	"
B10	M "	23	Open	"	"	"	"
B11	S0 "	5	25, 4, 9, 6	"	"	"	"
B12	S1 "	24	Open	"	"	"	"
B13	S2 "	16	8, 29, 30, 26, 40, 17, 36, 14	"	"	"	"
B14	S3 "	11	25, 4, 9, 6, 39, 18, 37, 31	"	"	"	"
B15	S0 "	34	39, 18, 37, 31	"	"	"	"
B16	S1 "	12	Open	"	"	"	"
B17	M "	15	Open	"	"	"	"

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TABLE VI. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B18	CN Input breakdown current	32	Open	0	1	mA	6
B19	B3 "	39	11, 34	"	"	"	"
B20	A3 "	40	11, 16, 39	"	"	"	"
B21	B2 "	18	11, 34	"	"	"	"
B22	A2 "	17	11, 16, 18	"	"	"	"
B23	B1 "	37	11, 34	"	"	"	"
B24	A1 "	36	11, 16, 37	"	"	"	"
B25	B0 "	31	11, 34	"	"	"	"
B26	A0 "	14	11, 16, 31	"	"	"	"

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CAUTION: Do not isolate the chasis of item X01 from earth ground.

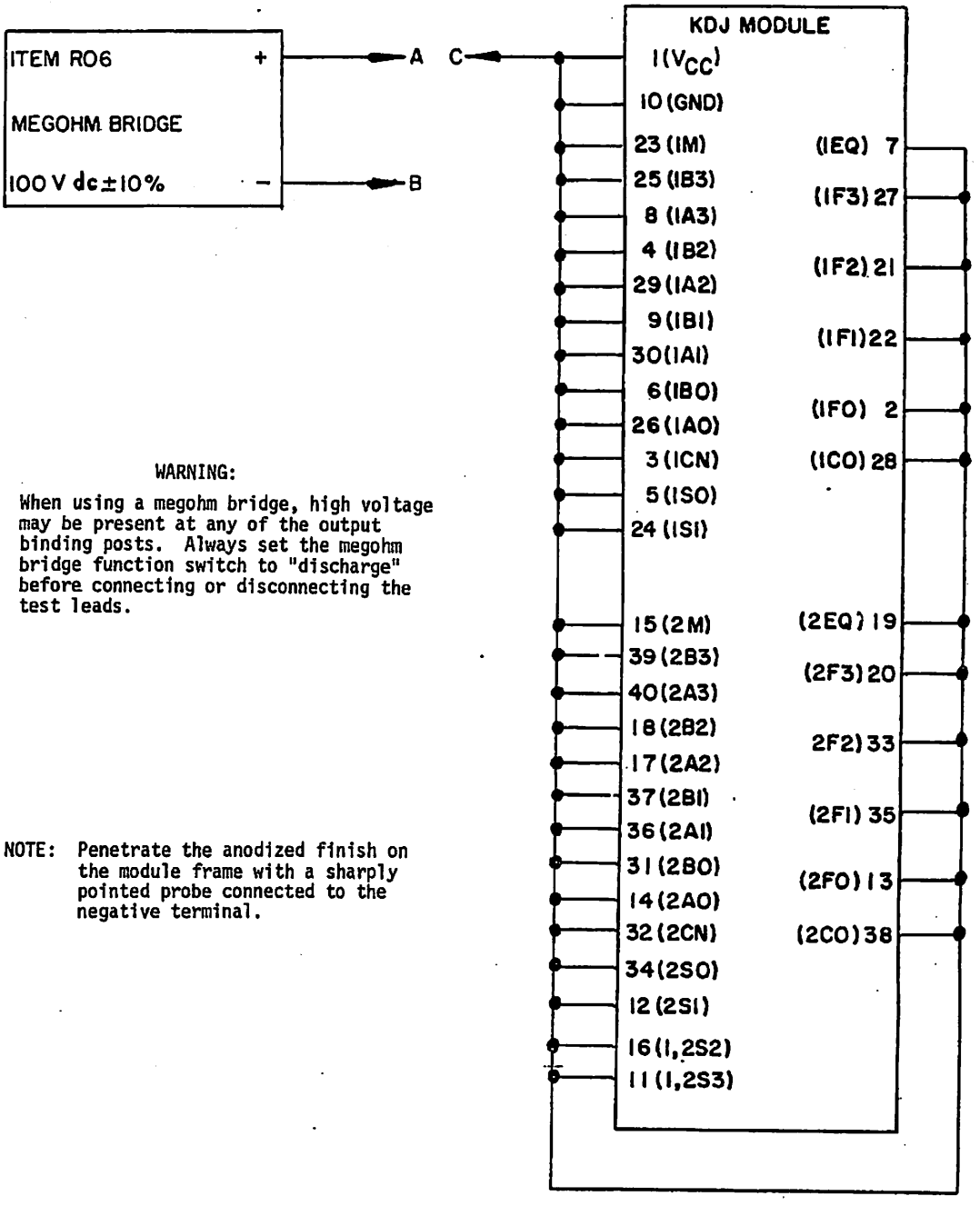
FIGURE 6. Input breakdown current test circuit for table VI.

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TABLE VII. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Isolation	C	Frame	10	∞	MΩ	7

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 7. Isolation test circuit for table VII.

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TABLE VIII. Input clamp diode voltage tests.

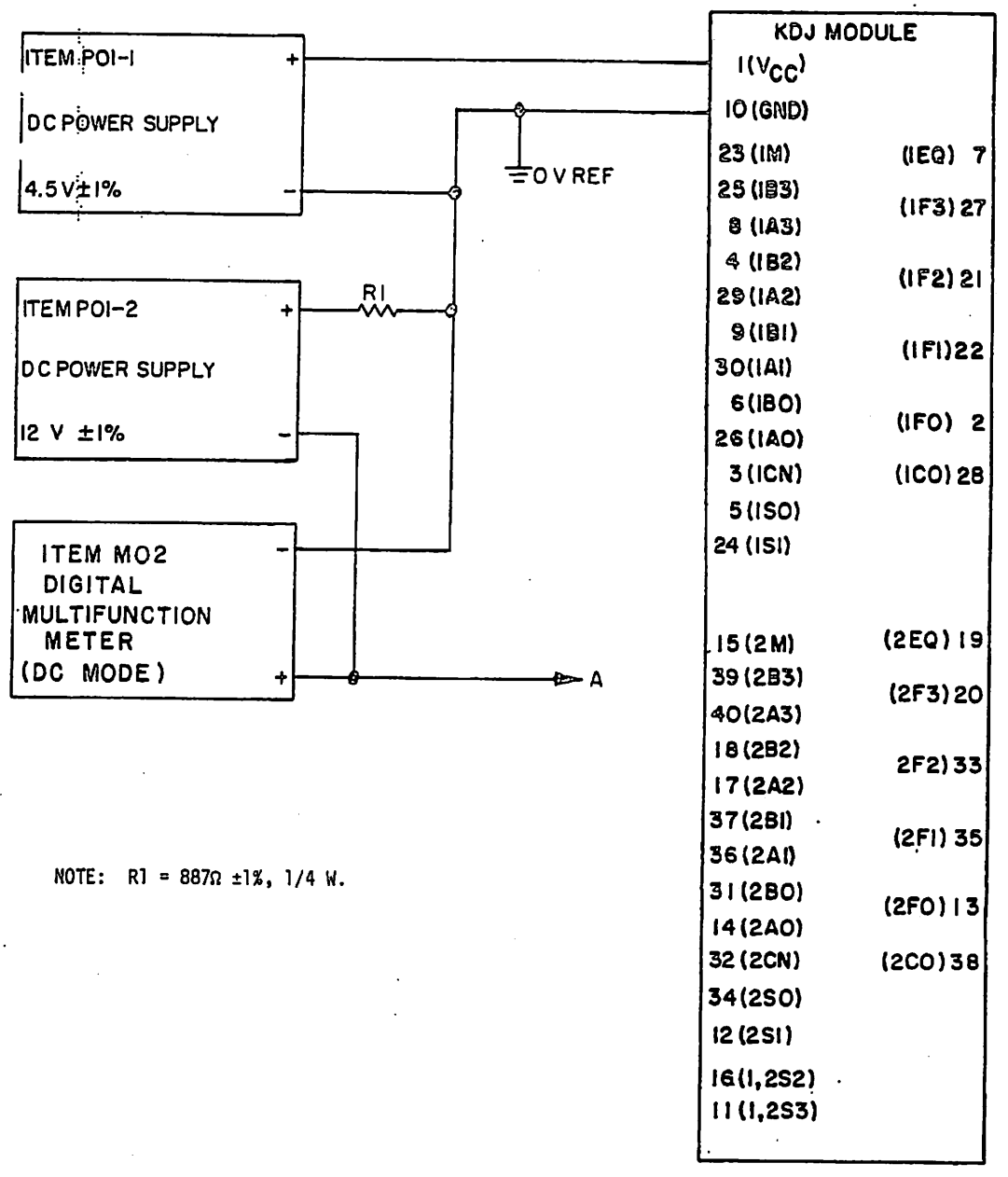
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D1	B3 Input clamp diode voltage	25	-0.4	-1.5	V	8
D2	A3 "	8	"	"	"	"
D3	B2 "	4	"	"	"	"
D4	A2 "	29	"	"	"	"
D5	B1 "	9	"	"	"	"
D6	A1 "	30	"	"	"	"
D7	B0 "	6	"	"	"	"
D8	A0 "	26	"	"	"	"
D9	CN "	3	"	"	"	"
D10	M "	23	"	"	"	"
D11	S0 "	5	"	"	"	"
D12	S1 "	24	"	"	"	"
D13	S2 "	16	"	"	"	"
D14	S3 "	11	"	"	"	"
D15	S0 "	34	"	"	"	"
D16	S1 "	12	"	"	"	"
D17	M "	15	"	"	"	"
D18	CN "	32	"	"	"	"

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TABLE VIII. Input clamp diode voltage tests - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D19	B3 Input clamp diode voltage	39	-0.4	-1.5	V	8
D20	A3 "	40	"	"	"	"
D21	B2 "	18	"	"	"	"
D22	A2 "	17	"	"	"	"
D23	B1 "	37	"	"	"	"
D24	A1 "	36	"	"	"	"
D25	B0 "	31	"	"	"	"
D26	A0 "	14	"	"	"	"

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NOTE: R1 = 887Ω ±1%, 1/4 W.

FIGURE 8. Input clamp diode voltage test circuit for table VIII.

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* TABLE IX. Output voltage (high-level) test.

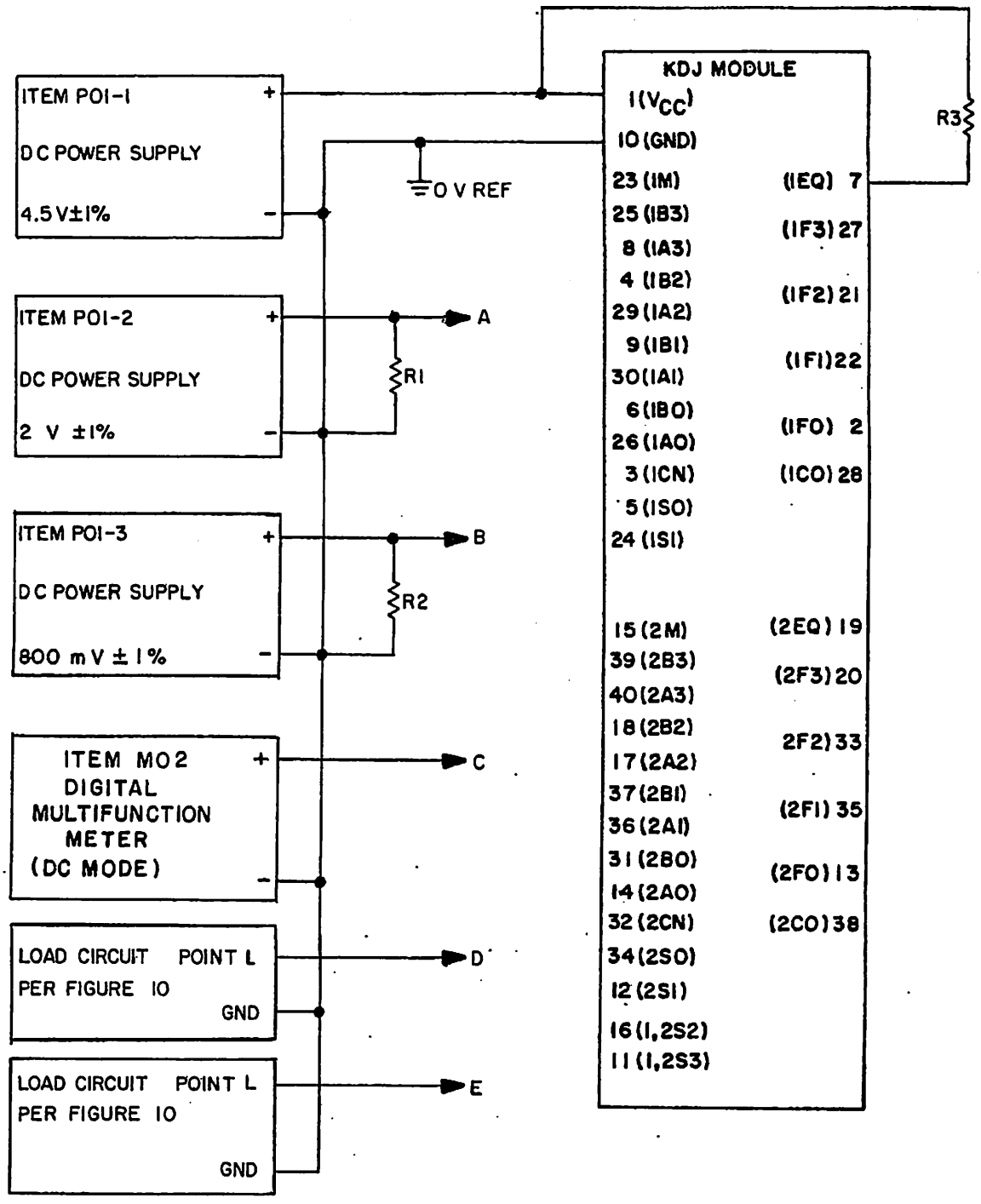
Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E1	F3 Output voltage, high level	8, 29, 30, 26, 40, 17, 36, 14, 5, 34, 11, 3, 32	25, 4, 9, 6, 39, 18, 37, 31, 12, 24, 16, 23, 15	27, D	27,C	Open	2.4	4.5	V	9
E2	F2 "	"	"	21, D	21, C	"	"	"	"	"
E3	F1 "	"	"	22, D	22, C	"	"	"	"	"
E4	F0 "	"	"	2, D	2, C	"	"	"	"	"
E5	C0 "	"	"	28, D	28, C	"	"	"	"	"
E6	C0 "	"	"	38, D	38, C	"	"	"	"	"
E7	F3 "	"	"	20, D	20, C	"	"	"	"	"
E8	F2 "	"	"	33, D	33, C	"	"	"	"	"
E9	F1 "	"	"	35, D	35, C	"	"	"	"	"
E10	F0 "	"	"	13, D	13, C	"	"	"	"	"
E11	EQ "	"	"	7, E	Open	7, C	"	"	"	"
E12	EQ "	"	"	19, E	Open	19, C	"	"	"	"

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* TABLE X. Output voltage (low-level) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F1	F3 Output voltage, low level	8, 29, 30, 26, 40, 17, 36, 14, 5, 34, 11	25, 4, 9, 6, 39, 18, 37, 31, 24, 16, 3, 23, 15, 32, 12	27, D	27,C	Open	0	400	mV	9
F2	F2 "	"	"	21, D	21, C	"	"	"	"	"
F3	F1 "	"	"	22, D	22, C	"	"	"	"	"
F4	F0 "	"	"	2, D	2, C	"	"	"	"	"
F5	C0 "	"	"	28, D	28, C	"	"	"	"	"
F6	E0 "	"	"	7, E	Open	7, C	"	"	"	"
F7	C0 "	"	"	38, D	38, C	Open	"	"	"	"
F8	E0 "	"	"	19, E	Open	19, C	"	"	"	"
F9	F3 "	"	"	20, D	20, C	Open	"	"	"	"
F10	F2 "	"	"	33, D	33, C	"	"	"	"	"
F11	F1 "	"	"	35, D	35, C	"	"	"	"	"
F12	F0 "	"	"	13, D	13, C	"	"	"	"	"

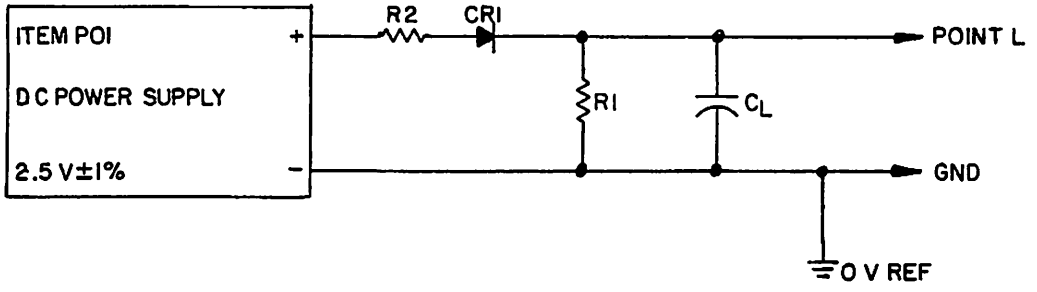
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 1.82 kΩ ±1%, 1/4 W.

* FIGURE 9. Output voltage (high-level and low-level) test circuit for tables IX and X.

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NOTES:
1. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions L and M.
2. CR1 = 1N4150 or equivalent.
3. R1 = 3.01 kΩ ±1%, 1/8 W.
4. R2 = Select value for current flow of 16 mA ±1% from point L to ground with point L held at 400 mV for all outputs except 2EQ (pin 19) and 1EQ (pin 7).
5. R2 = Select value for current flow of 12.8 mA ±1% from point L to ground with point L held at 400 mV for 2EQ (pin 19) and 1EQ (pin 7) outputs.
6. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test condition R.
7. Capacitor C_L is not required for dc electrical tests E1-E12 or F1-F12.
8. Lead length from load circuit to module output pin under test is to be less than 3 inches (76.2 mm).
9. During output switching, power supply item POI maximum overshoot shall be less than 500 mV.

* FIGURE 10. KDJ logic load circuit.

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TABLE XI. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	M1 Input current, low level	Open	23	Open	0.40	1.73	mA	11
G2	M2 "	"	15	"	0.40	1.73	"	"
G3	B3 "	"	25	"	1.20	5.17	"	"
G4	A3 "	"	8	"	"	"	"	"
G5	B2 "	"	4	"	"	"	"	"
G6	A2 "	"	29	"	"	"	"	"
G7	B1 "	"	9	"	"	"	"	"
G8	A1 "	"	30	"	"	"	"	"
G9	B0 "	"	6	"	"	"	"	"
G10	A0 "	"	26	"	"	"	"	"
G11	B3 "	"	39	"	"	"	"	"
G12	A3 "	"	40	"	"	"	"	"
G13	B2 "	"	18	"	"	"	"	"
G14	A2 "	"	17	"	"	"	"	"
G15	B1 "	"	37	"	"	"	"	"
G16	A1 "	"	36	"	"	"	"	"
G17	B0 "	"	31	"	"	"	"	"
G18	A0 "	"	14	"	"	"	"	"
G19	S0 "	"	5	"	1.6	6.9	"	"

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TABLE XI. Input current (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G20	S1 Input current, low level	Open	24	Open	1.6	6.9	mA	11
G21	SO "	"	34	"	"	"	"	"
G22	S1 "	"	12	"	"	"	"	"
G23	S2 "	"	16	"	3.2	13.8	"	"
G24	S3 "	"	11	"	3.2	13.8	"	"
G25	CN "	"	3	23, 5, 24, 16, 11, 26, 30, 29, 8	2.00	8.62	"	"
G26	CN "	"	32	34, 12, 16, 11, 15, 14, 36, 17, 40	2.00	8.62	"	"

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* TABLE XII. Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	M1 Input current, high level	23	Open	Open	0	40	µA	11
H2	M2 "	15	"	"	"	40	"	"
H3	B3 "	25	"	"	"	120	"	"
H4	A3 "	8	"	"	"	"	"	"
H5	B2 "	4	"	"	"	"	"	"
H6	A2 "	29	"	"	"	"	"	"
H7	B1 "	9	"	"	"	"	"	"
H8	A1 "	30	"	"	"	"	"	"
H9	B0 "	6	"	"	"	"	"	"
H10	A0 "	26	"	"	"	"	"	"
H11	B3 "	39	"	"	"	"	"	"
H12	A3 "	40	"	"	"	"	"	"
H13	B2 "	18	"	"	"	"	"	"
H14	A2 "	17	"	"	"	"	"	"
H15	B1 "	37	"	"	"	"	"	"
H16	A1 "	36	"	"	"	"	"	"
H17	B0 "	31	"	"	"	"	"	"
H18	A0 "	14	"	"	"	"	"	"
H19	S0 "	5	"	"	"	160	"	"

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* TABLE XII. Input current (high-level) test - Continued.

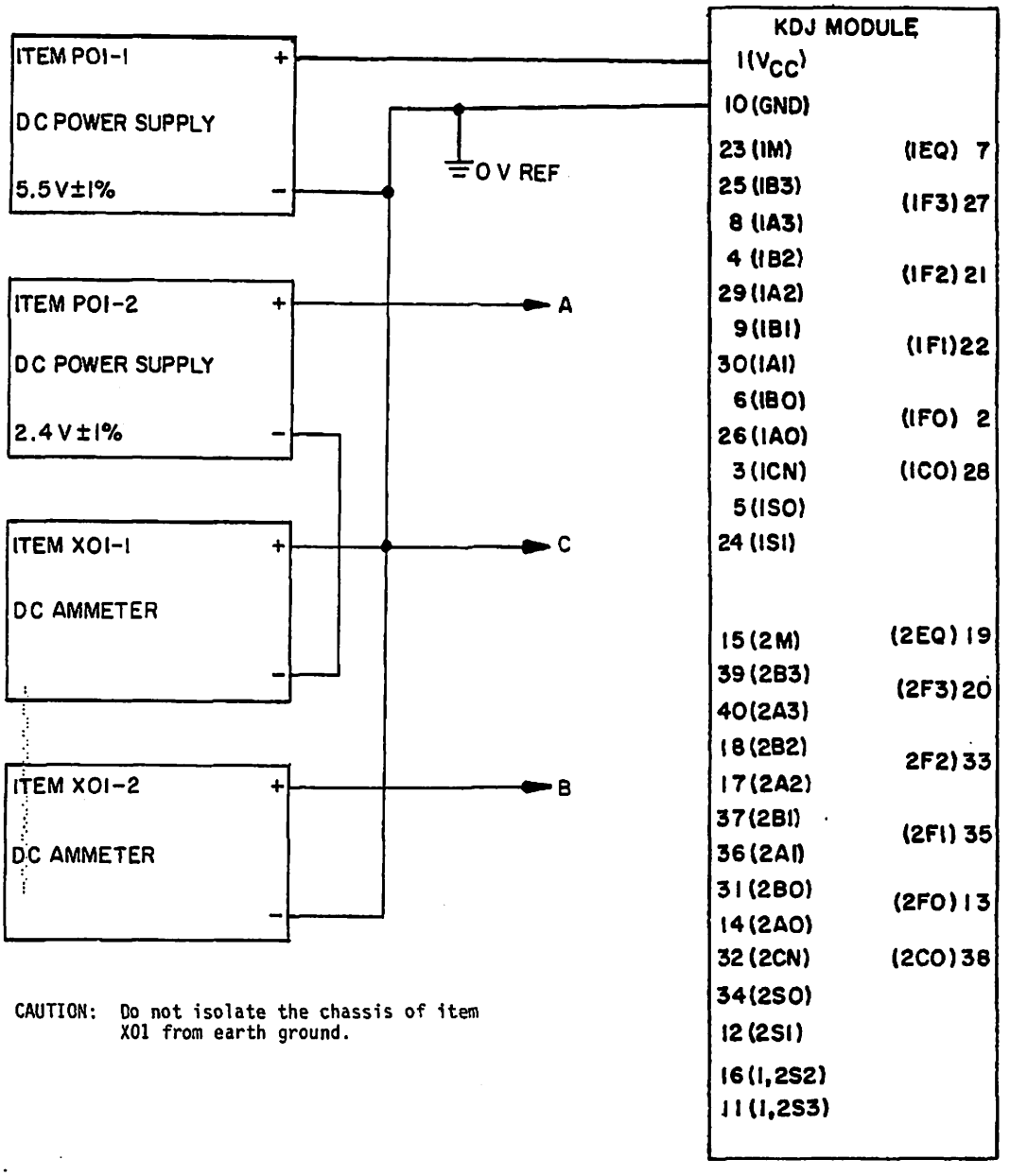
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H20	S1 Input current, high level	24	Open	Open	0	160	µA	11
H21	S0 "	34	"	"	"	"	"	"
H22	S1 "	12	"	"	"	"	"	"
H23	S2 "	16	"	"	"	320	"	"
H24	S3 "	11	"	"	"	320	"	"
H25	CN "	3	"	"	"	200	"	"
H26	CN "	32	"	"	"	200	"	"

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* TABLE XIII. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	F3 Output short circuit current	8,29, 30,26, 40, 17, 36, 14, 5, 34, 11, 3, 32	27	25, 4, 9, 6, 39, 18, 37, 31, 12, 24, 16, 23, 15	20	55	mA	11
J2	F2 "	"	21	"	"	"	"	"
J3	F1 "	"	22	"	"	"	"	"
J4	F0 "	"	2	"	"	"	"	"
J5	C0 "	"	28	"	"	"	"	"
J6	C0 "	"	38	"	"	"	"	"
J7	F3 "	"	20	"	"	"	"	"
J8	F2 "	"	33	"	"	"	"	"
J9	F1 "	"	35	"	"	"	"	"
J10	F0 "	"	13	"	"	"	"	"
J11	EQ "	"	19	"	2.4	3.4	"	"
J12	EQ "	"	7	"	0	500	µA	"

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CAUTION: Do not isolate the chassis of item XO1 from earth ground.

NOTES:
1. Since items XO1-1 and XO1-2 are not used simultaneously, only one item XO1 is required.
2. With point A disconnected, item XO1-1 should indicate zero current on the 1 µA range.
3. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

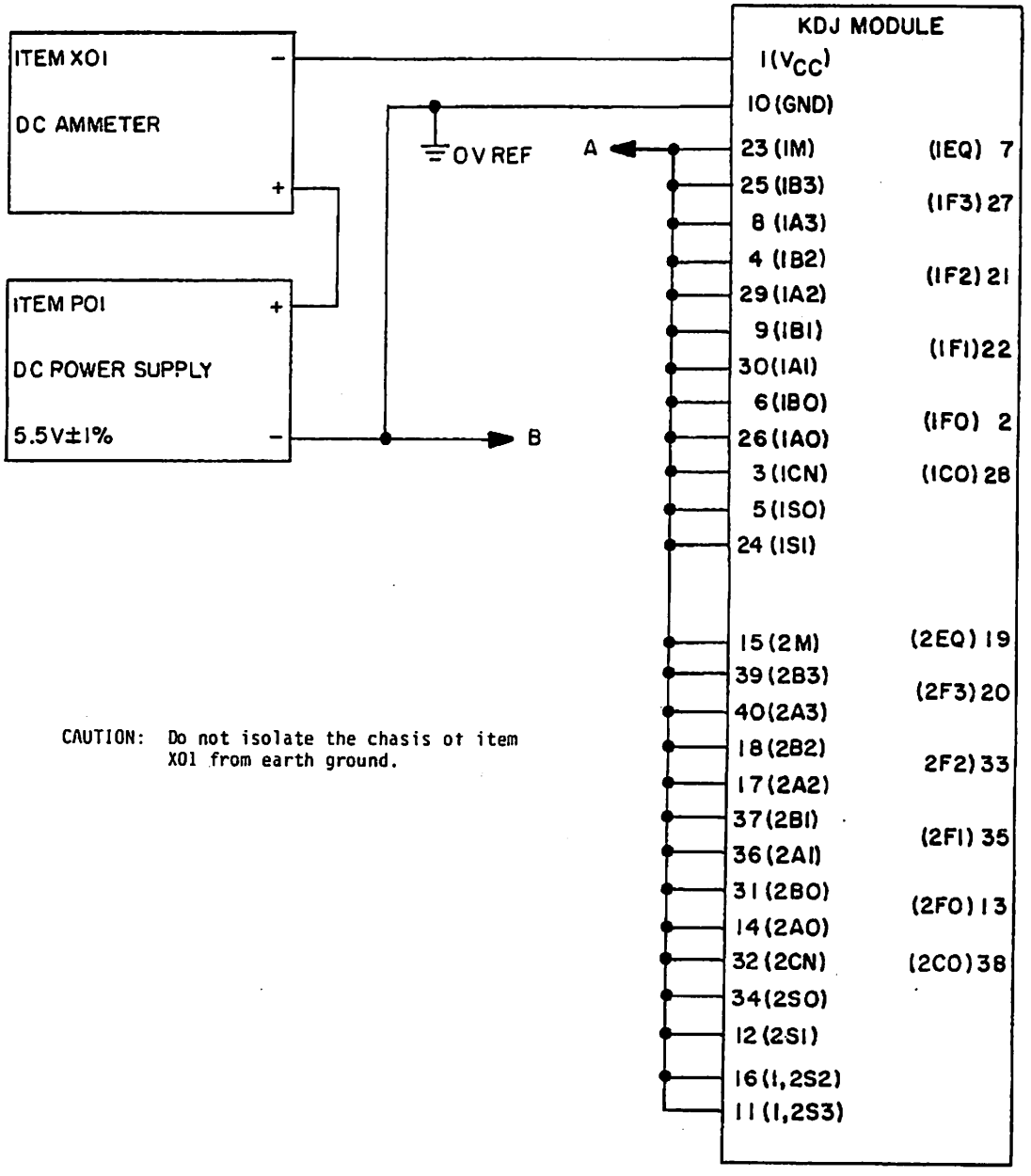
* FIGURE 11. Input current (low-level and high-level) and output short circuit current test circuit for tables XI, XII, and XIII.

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* TABLE VIV. Power supply current (high-level and low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Power supply current, high level	Open	Open	1	254	mA	12
K2	Power supply current, low level	B	A	1	273	mA	12

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CAUTION: Do not isolate the chasis of item XO1 from earth ground.

* FIGURE 12. Power supply current (high-level and low-level) test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	D to pin	B to pin	C to pin	Min	Max		
L1	Transition time (t_{TLH})	8, 29, 30, 26, 40, 17, 36, 14, 5, 34, 11	25, 4, 9, 6, 39, 18, 37, 31, 24, 12, 16, 23, 15	3, 32	27	4	45	ns	13
L2	"	"	"	"	21	"	"	"	"
L3	"	"	"	"	22	"	"	"	"
L4	"	"	"	"	2	"	"	"	"
L5	"	"	"	"	28	"	"	"	"
L6	"	"	"	"	38	"	"	"	"
L7	"	"	"	"	20	"	"	"	"
L8	"	"	"	"	33	"	"	"	"
L9	"	"	"	"	35	"	"	"	"
L10	"	"	"	"	13	"	"	"	"
L11	"	"	"	"	7	5	"	"	"
L12	"	"	"	"	19	5	"	"	"
L13	Transition time (t_{THL})	"	"	"	27	4	20	"	"

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	D to pin	B to pin	C to pin	Min	Max		
L14	Transition time (t_{THL})	8, 29, 30, 26, 40, 17, 36, 14, 5, 34, 11	25, 4, 9, 6, 39, 18, 37, 31, 24, 12, 16, 23, 15	3, 32	21	4	20	ns	13
L15	"	"	"	"	22	"	"	"	"
L16	"	"	"	"	2	"	"	"	"
L17	"	"	"	"	28	"	"	"	"
L18	"	"	"	"	7	1	"	"	"
L19	"	"	"	"	38	4	"	"	"
L20	"	"	"	"	19	1	"	"	"
L21	"	"	"	"	20	4	"	"	"
L22	"	"	"	"	33	"	"	"	"
L23	"	"	"	"	35	"	"	"	"
L24	"	"	"	"	13	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	D to pin	B to pin	C to pin	Min	Max		
M1	A to F Propagation delay time (t_{PLH})	5, 34, 11, 3, 32, 30, 29, 8, 6	24, 12, 16, 23, 15, 9, 4, 25	26	2	6	41	ns	13
M2	A to F Propagation delay time (t_{PHL})	5, 34, 11, 3, 32, 30, 29, 8, 6	24, 12, 16, 23, 15, 9, 4, 25	26	2	"	50	"	"
M3	A to F Propagation delay time (t_{PLH})	5, 34, 11, 3, 32, 26, 29, 8, 9	24, 12, 16, 23, 15, 6, 4, 25	30	22	"	41	"	"
M4	A to F Propagation delay time (t_{PHL})	5, 34, 11, 3, 32, 26, 29, 8, 9	24, 12, 16, 23, 15, 6, 4, 25	30	22	"	50	"	"
M5	A to F Propagation delay time (t_{PLH})	5, 34, 11, 3, 32, 26, 30, 8, 4	24, 12, 16, 23, 15, 6, 9, 25	29	21	"	41	"	"
M6	A to F Propagation delay time (t_{PHL})	5, 34, 11, 3, 32, 26, 30, 8, 4	24, 12, 16, 23, 15, 6, 9, 25	29	21	"	50	"	"
M7	A to F Propagation delay time (t_{PLH})	5, 34, 11, 3, 32, 26,	24, 12, 16, 23, 15, 6,	8	27	"	41	"	"
M8	A to F Propagation delay time (t_{PHL})	5, 34, 11, 3, 32, 26, 30, 29, 25	24, 12, 16, 23, 15, 6, 9, 4	8	27	"	50	"	"
M9	A to F Propagation delay time (t_{PLH})	5, 34, 11, 3, 32, 36, 17, 40, 31	24, 12, 16, 23, 15, 37, 18, 39	14	13	"	41	"	"
M10	A to F Propagation delay time (t_{PHL})	5, 34, 11, 3, 32, 36, 17, 40, 31 30, 29, 25	24, 12, 16, 23, 15, 37, 18, 39 9, 4	14	13	"	50	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	D to pin	B to pin	C to pin	Min	Max		
M11	A to F Propagation delay time (t_{PLH})	5, 34, 11, 3, 32, 14, 17, 40, 37	24, 12, 16, 23, 15, 31, 18, 39	36	35	6	41	ns	13
M12	A to F Propagation delay time (t_{PHL})	5, 34, 11, 3, 32, 14, 17, 40, 37	24, 12, 16, 23, 15, 31, 18, 39	36	35	"	50	"	"
M13	A to F Propagation delay time (t_{PLH})	5, 34, 11, 3, 32, 14, 36, 40, 18	24, 12, 16, 23, 15, 31, 37, 39	17	33	"	41	"	"
M14	A to F Propagation delay time (t_{PHL})	5, 34, 11, 3, 32, 14, 36, 40, 18	24, 12, 16, 23, 15, 31, 37, 39	17	33	"	50	"	"
M15	A to F Propagation delay time (t_{PLH})	5, 34, 11, 3, 32, 14, 36, 17, 39	24, 12, 16, 23, 15, 31, 37, 18	40	20	"	41	"	"
M16	A to F Propagation delay time (t_{PHL})	5, 34, 11, 3, 32, 14, 36, 17, 39	24, 12, 16, 23, 15, 31, 37, 18	40	20	"	50	"	"
M17	B to F Propagation delay time (t_{PLH})	5, 34, 11, 3, 32	24, 12, 16, 23, 15, 8, 29, 30, 26	6	2	"	50	"	"
M18	B to F Propagation delay time (t_{PHL})	"	"	6	2	"	41	"	"
M19	B to F Propagation delay time (t_{PLH})	"	"	9	22	"	50	"	"
M20	B to F Propagation delay time (t_{PHL})	"	"	9	22	"	41	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	D to pin	B to pin	C to pin	Min	Max		
M21	B to F Propagation delay time (t_{PLH})	5, 34, 11, 3, 32	24, 12, 16, 23, 15, 8, 29, 30, 26	4	21	6	50	ns	13
M22	B to F Propagation delay time (t_{PHL})	"	"	4	21	"	41	"	"
M23	B to F Propagation delay time (t_{PLH})	"	"	25	27	"	50	"	"
M24	B to F Propagation delay time (t_{PHL})	"	"	25	27	"	41	"	"
M25	B to F Propagation delay time (t_{PLH})	"	24, 12, 16, 23, 15, 40, 17, 36, 14	31	13	"	50	"	"
M26	B to F Propagation delay time (t_{PHL})	"	"	31	13	"	41	"	"
M27	B to F Propagation delay time (t_{PLH})	"	"	37	35	"	50	"	"
M28	B to F Propagation delay time (t_{PHL})	"	"	37	35	"	41	"	"
M29	B to F Propagation delay time (t_{PLH})	"	"	18	33	"	50	"	"
M30	B to F Propagation delay time (t_{PHL})	"	"	18	33	"	41	"	"
M31	B to F Propagation delay time (t_{PLH})	"	"	39	20	"	50	"	"
M32	B to F Propagation delay time (t_{PHL})	"	"	39	20	"	41	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	D to pin	B to pin	C to pin	Min	Max		
M33	CN to F and CN to CO Propagation delay time (t_{PLH})	8, 29, 30, 26, 40, 17, 36, 14, 5, 34, 11	25, 4, 9, 6, 39, 18, 37, 31, 24, 12, 16, 23, 15	3	27	4	27	ns	13
M34	CN to F and CN to CO Propagation delay time (t_{PHL})	8, 29, 30, 26, 40, 17, 36, 14, 5, 34, 11	25, 4, 9, 6, 39, 18, 37, 31, 24, 12, 16, 23, 15	"	21	"	"	"	"
M35	CN to F and CN to CO Propagation delay time (t_{PLH})	"	"	"	22	"	"	"	"
M36	"	"	"	"	2	"	"	"	"
M37	"	"	"	32	20	"	"	"	"
M38	"	"	"	"	33	"	"	"	"
M39	"	"	"	"	35	"	"	"	"
M40	"	"	"	"	13	"	"	"	"
M41	"	"	"	"	38	"	26	"	"
M42	"	"	"	3	28	"	26	"	"
M43	CN to F and CN to CO Propagation delay time (t_{PHL})	"	"	3	28	"	28	"	"
M44	"	"	"	32	38	"	28	"	"
M45	"	"	"	"	20	"	27	"	"
M46	"	"	"	"	33	"	27	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	D to pin	B to pin	C to pin	Min	Max		
M47	CN to F and CN to CO Propagation delay time (t_{PLH})	8, 29, 30, 26, 40, 17, 36, 14, 5, 34, 11	25, 4, 9, 6, 39, 18, 37, 31, 24, 12, 16, 23, 15	32	35	4	27	ns	13
M48	"	"	"	32	13	"	"	"	"
M49	"	"	"	3	27	"	"	"	"
M50	"	"	"	"	21	"	"	"	"
M51	"	"	"	"	22	"	"	"	"
M52	"	"	"	"	2	"	"	"	"
M53	A to EQ Propagation delay time (t_{PLH})	12, 24, 16, 3, 32, 25, 4, 9, 6, 30, 29, 8	5, 34, 11, 23, 15	26	7	6	63	"	"
M54	A to EQ Propagation delay time (t_{PHL})	12, 24, 16, 3, 32, 25, 4, 9, 6, 30, 29, 8	"	26	"	"	"	"	"
M55	A to EQ Propagation delay time (t_{PLH})	12, 24, 16, 3, 32, 25, 4, 9, 6, 26, 29, 8	"	30	"	"	"	"	"
M56	A to EQ Propagation delay time (t_{PHL})	12, 24, 16, 3, 32, 25, 4, 9, 6, 26, 29, 8	"	30	"	"	"	"	"
M57	A to EQ Propagation delay time (t_{PLH})	12, 24, 16, 3, 32, 25, 4, 9, 6, 26, 30, 8	"	29	"	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	D to pin	B to pin	C to pin	Min	Max		
M58	A to EQ Propagation delay time (t_{PHL})	12, 24, 16, 3, 32, 25, 4, 9, 6, 26, 30, 8	5, 34, 11, 23, 15	29	7	6	63	ns	13
M59	A to EQ Propagation delay time (t_{PLH})	12, 24, 16, 3, 32, 25, 4, 9, 6, 26, 30, 29	"	8	"	"	"	"	"
M60	A to EQ Propagation delay time (t_{PHL})	12, 24, 16, 3, 32, 25, 4, 9, 6, 26, 30, 29	"	8	"	"	"	"	"
M61	A to EQ Propagation delay time (t_{PLH})	12, 24, 16, 3, 32, 39, 18, 37, 31, 36, 17, 40	"	14	19	"	"	"	"
M62	A to EQ Propagation delay time (t_{PHL})	12, 24, 16, 3, 32, 39, 18, 37, 31, 36, 17, 40	"	14	"	"	"	"	"
M63	A to EQ Propagation delay time (t_{PLH})	12, 24, 16, 3, 32, 39, 18, 37, 31, 14, 17, 40	"	36	"	"	"	"	"
M64	A to EQ Propagation delay time (t_{PHL})	12, 24, 16, 3, 32, 39, 18, 37, 31, 14, 17, 40	"	36	"	"	"	"	"
M65	A to EQ Propagation delay time (t_{PLH})	12, 24, 16, 3, 32, 39, 18, 37, 31, 14, 36, 40	"	17	"	"	"	"	"
M66	A to EQ Propagation delay time (t_{PHL})	12, 24, 16 3, 32, 39, 18, 37, 31 14, 36, 40	"	17	"	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

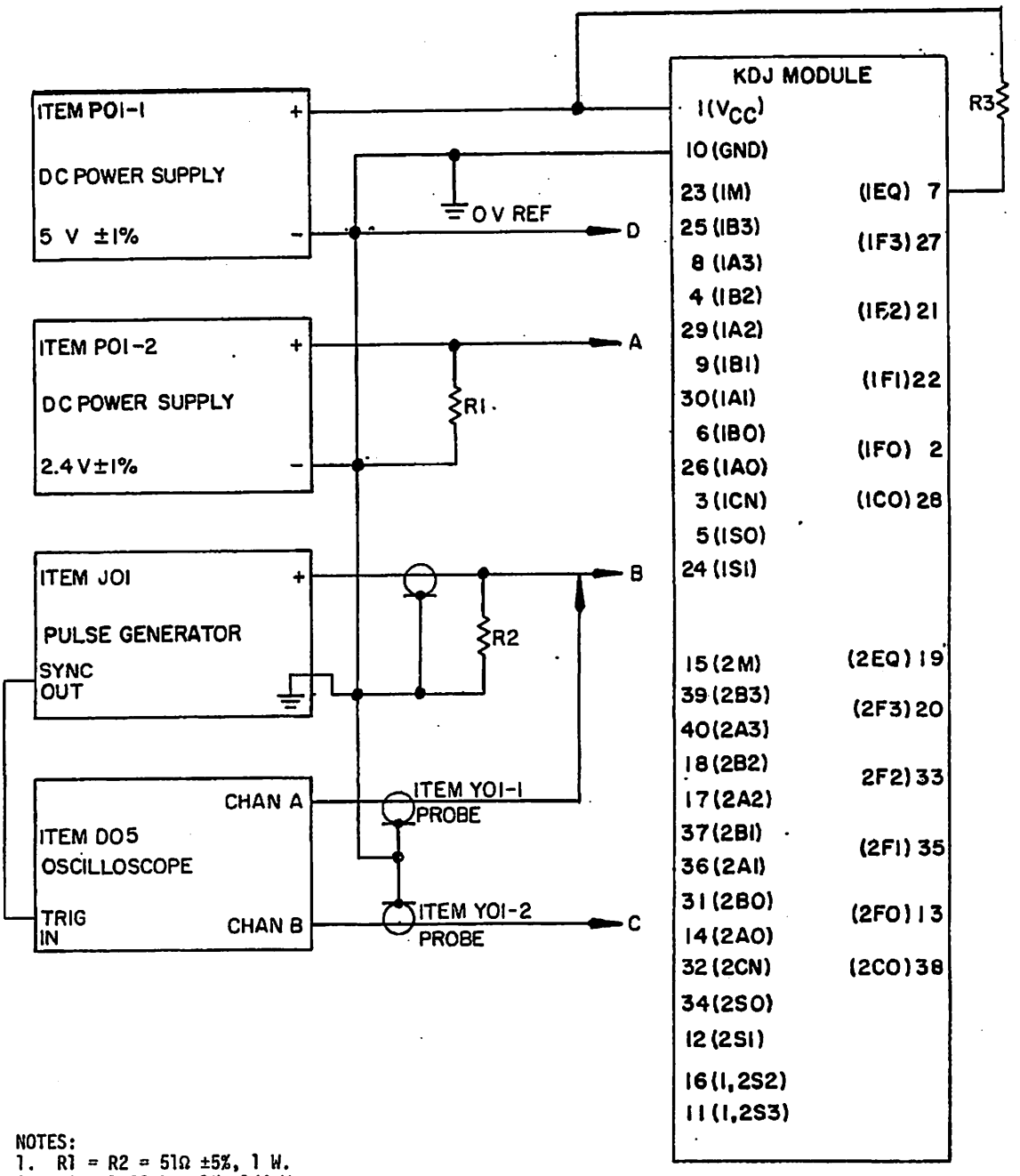
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	D to pin	B to pin	C to pin	Min	Max		
M67	A to EQ Propagation delay time (t_{PLH})	12, 24, 16, 3, 32, 39, 18, 37, 31, 14, 36, 17	5, 34, 11, 23, 15	40	19	6	63	ns	13
M68	A to EQ Propagation delay time (t_{PHL})	12, 24, 16, 3, 32, 39, 18, 37, 31, 14, 36, 17	"	40	"	"	"	"	"
M69	B to EQ Propagation delay time (t_{PLH})	12, 24, 16, 3, 32	5, 34, 11, 23, 15, 8, 29, 30, 26, 9, 4, 25	6	7	"	"	"	"
M70	B to EQ Propagation delay time (t_{PHL})	"	5, 34, 11, 23, 15, 8, 29, 30, 26, 9, 4, 25	6	"	"	"	"	"
M71	B to EQ Propagation delay time (t_{PLH})	"	5, 34, 11, 23, 15, 8, 29, 30, 26, 6, 4, 25	9	"	"	"	"	"
M72	B to EQ Propagation delay time (t_{PHL})	"	5, 34, 11, 23, 15, 8, 29, 30, 26, 6, 4, 25	9	"	"	"	"	"
M73	B to EQ Propagation delay time (t_{PLH})	"	5, 34, 11, 23, 15, 8, 29, 30, 26, 6, 9, 25	4	"	"	"	"	"
M74	B to EQ Propagation delay time (t_{PHL})	"	5, 34, 11, 23, 15, 8, 29, 30, 26, 6, 9, 25	4	"	"	"	"	"
M75	B to EQ Propagation delay time (t_{PLH})	"	5, 34, 11, 23, 15, 8, 29, 30, 26, 6, 9, 4	25	"	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	D to pin	B to pin	C to pin	Min	Max		
M76	B to EQ Propagation delay time (t_{PHL})	12, 24, 16, 3, 32	5, 34, 11, 23, 15, 8, 29, 30, 26, 6, 9, 4	25	7	6	63	ns	13
M77	B to EQ Propagation delay time (t_{PLH})	"	5, 34, 11, 23, 15, 40, 17, 36, 14, 37, 18, 39	31	19	"	"	"	"
M78	B to EQ Propagation delay time (t_{PHL})	"	5, 34, 11, 23, 15, 40, 17, 36, 14, 37, 18, 39	31	"	"	"	"	"
M79	B to EQ Propagation delay time (t_{PLH})	"	5, 34, 11, 23, 15, 40, 17, 36, 14, 31, 18, 39	37	"	"	"	"	"
M80	B to EQ Propagation delay time (t_{PHL})	"	5, 34, 11, 23, 15, 40, 17, 36, 14, 31, 18, 39	37	"	"	"	"	"
M81	B to EQ Propagation delay time (t_{PLH})	"	5, 34, 11, 23, 15, 40, 17, 36, 14, 31, 37, 39	18	"	"	"	"	"
M82	B to EQ Propagation delay time (t_{PHL})	"	5, 34, 11, 23, 15, 40, 17, 36, 14, 31, 37, 39	18	"	"	"	"	"
M83	B to EQ Propagation delay time (t_{PLH})	"	5, 34, 11, 23, 15, 40, 17, 36, 14, 31, 37, 18	39	"	"	"	"	"
M84	B to EQ Propagation delay time (t_{PHL})	"	5, 34, 11, 23, 15, 40, 17, 36, 14, 31, 37, 18	39	"	"	"	"	"

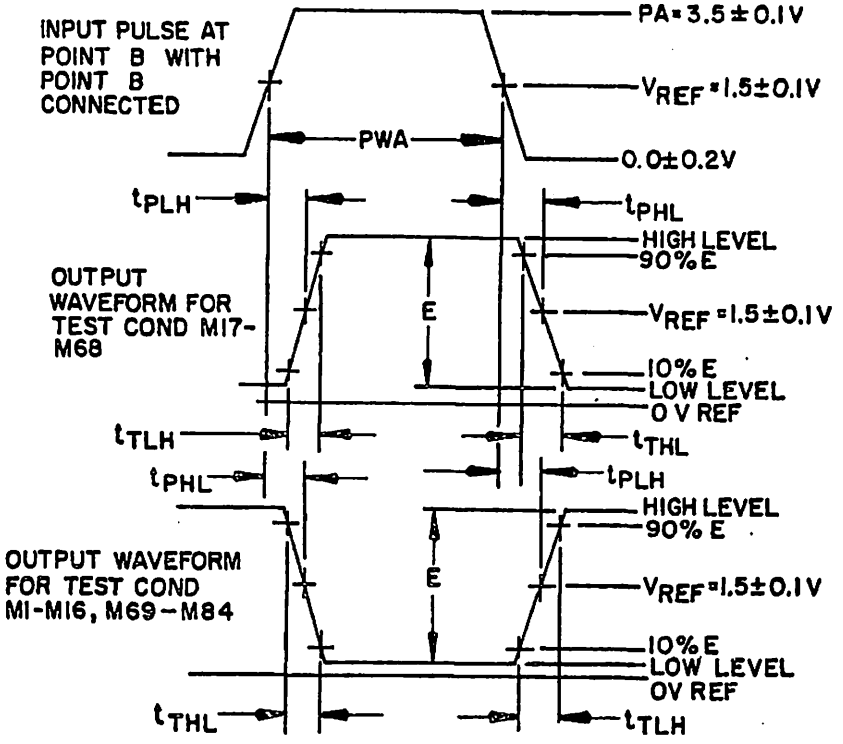
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 1.82 kΩ ±1%, 1/4 W.
3. Load output under test with a load as specified on figure 10.
4. Before making measurements specified in tables XV and XVI, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.4 V.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XV and XVI.

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NOTES:

1. Input pulse parameters for test conditions L1-L10, L13-L17, L19, L21-L24, and M1-M52.
   a. PA = 3.5 ±0.1 V.
   b. PWA = 300 ns ±10%.
   c. $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   d. PRR = 1 MHz ±10%.
   e. All voltage levels are reference to ground.
2. Input pulse parameters for test conditions L11, L12, L18, L20, and M53-M84.
   a. PA = 3.5 ±0.1 V.
   b. PWA = 1.0 ±0.1 µs.
   c. $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   d. PRR = 500 kHz ±10%.
   e. For L11, L12, L18, and L20 EQ output t_{TLH} and t_{THL} shall be measured between the 10% E and 50% E points.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XV and XVI- Continued.

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TABLE XVII. Input capacitance test.

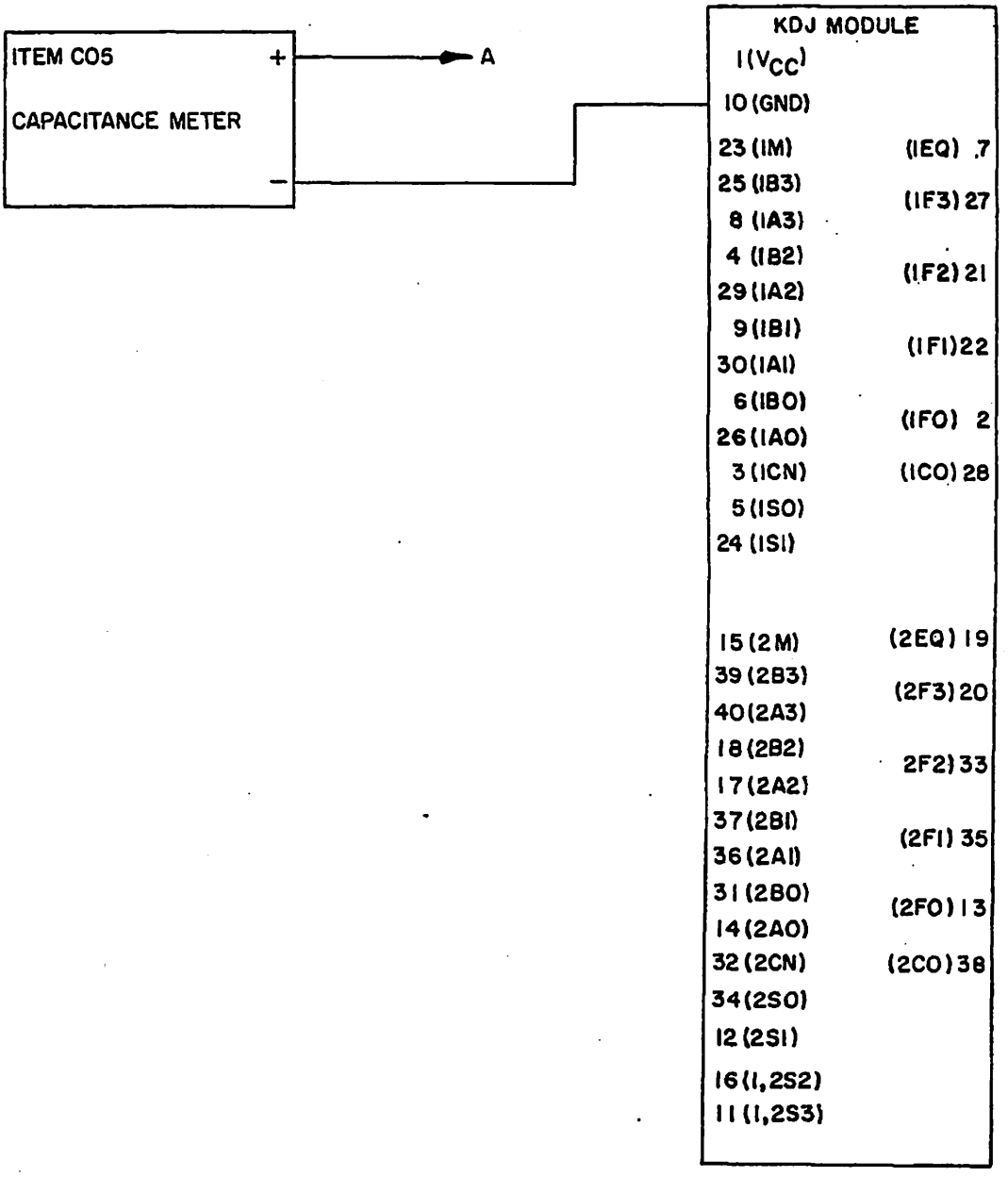
Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
N1	B3 Input capacitance	25	2	18	pF	14
N2	A3 "	8	"	"	"	"
N3	B2 "	4	"	"	"	"
N4	A2 "	29	"	"	"	"
N5	B1 "	9	"	"	"	"
N6	A1 "	30	"	"	"	"
N7	B0 "	6	"	"	"	"
N8	A0 "	26	"	"	"	"
N9	CN "	3	"	"	"	"
N10	M "	23	"	"	"	"
N11	S0 "	5	"	"	"	"
N12	S1 "	24	"	"	"	"
N13	S0 "	34	"	"	"	"
N14	S1 "	12	"	"	"	"
N15	M "	15	"	"	"	"
N16	CN "	32	"	"	"	"
N17	B3 "	39	"	"	"	"
N18	A3 "	40	"	"	"	"

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TABLE XVII. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
N19	B2 Input capacitance	18	2	18	pF	14
N20	A2 "	17	"	"	"	"
N21	B1 "	37	"	"	"	"
N22	A1 "	36	"	"	"	"
N23	B0 "	31	"	"	"	"
N24	A0 "	14	"	"	"	"
N25	S2 "	16	"	35	"	"
N26	S3 "	11	"	35	"	"

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

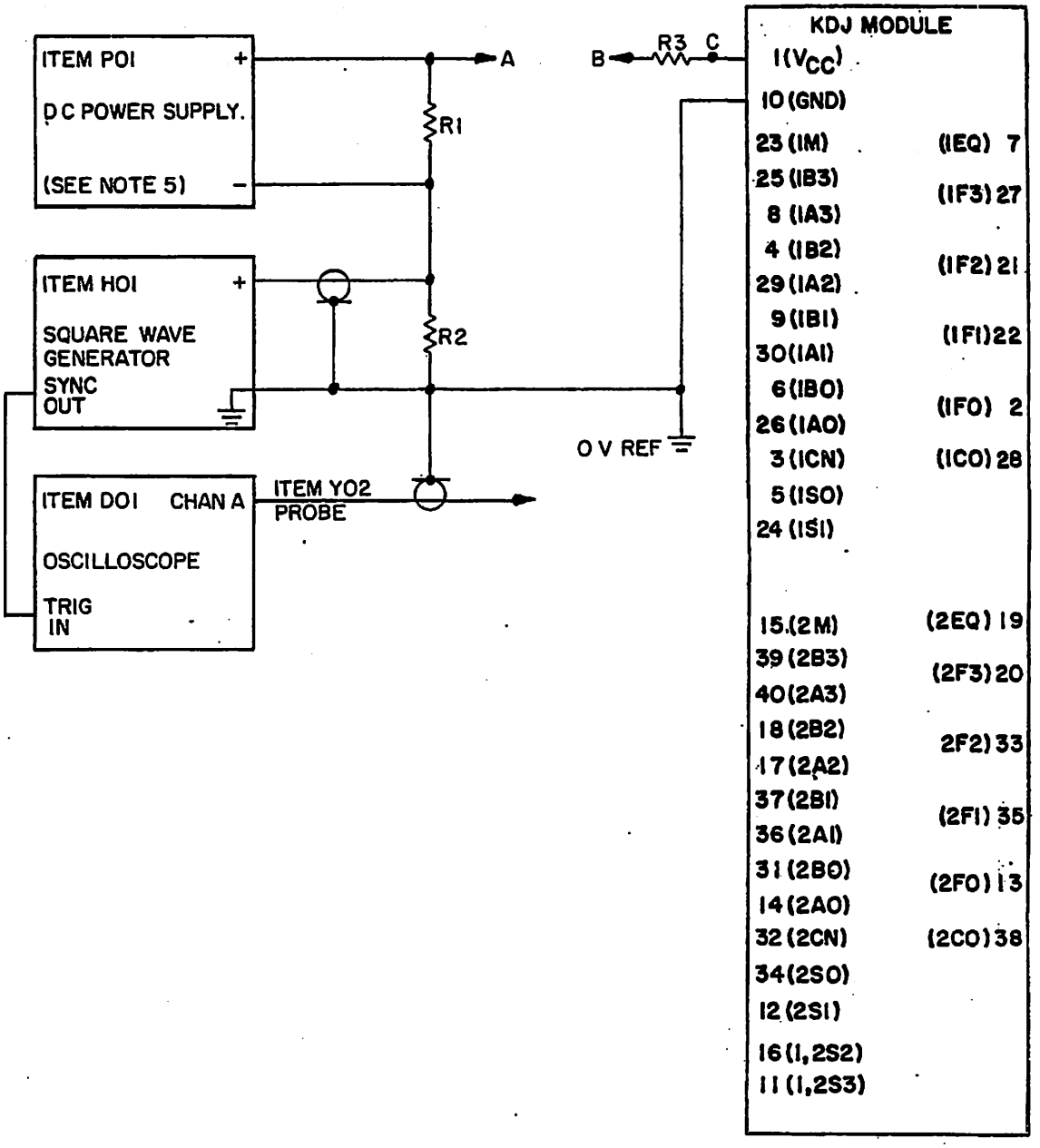
FIGURE 14. Input capacitance test circuit for table XVII.

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TABLE XVIII. Power supply transient rise time.

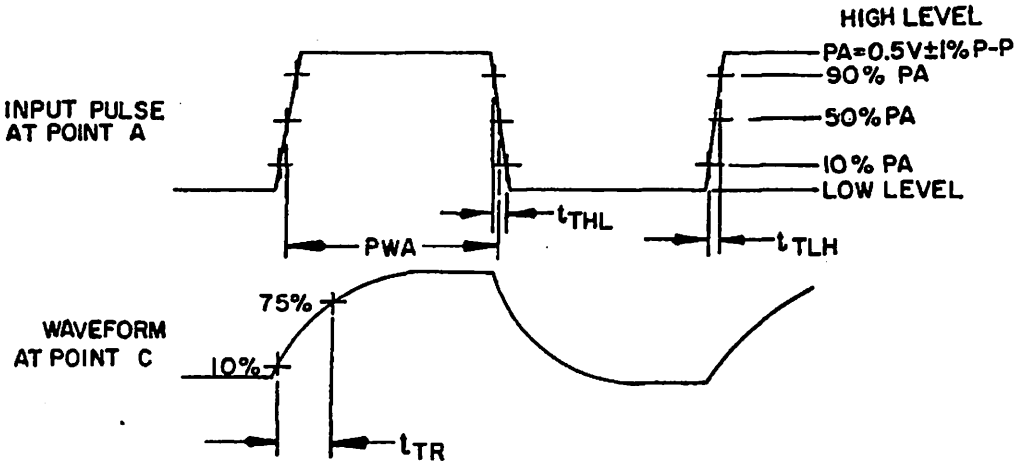
Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
P1	Power supply transient rise time	1	2	ms	15

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* FIGURE 15. Power supply transient rise time test circuit and waveforms for table XVIII.

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NOTES:
1. PWA = 10 ms ±10%.
2. PRR = 50 Hz (reference).
3. t_{TLH} and t_{THL} < 100 ns (reference).
4. With point A open, adjust input as shown.
5. With point A open, adjust item P01 so that the low level of the input is at 0 ±0.01 V.
6. Connect A to B; recheck the low level and pulse parameters at point A.
7. Measure transient rise time (t_{TR}) at point C.
8. R1 = R2 = 51Ω ±5%, 1 W.
9. R3 = 511Ω ±1%, 1/4 W.

* FIGURE 15. Power supply transient rise time test circuit and waveforms for table XVIII - Continued.

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* TABLE XIX. Output crosstalk (high-level and low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	Output crosstalk, high level	5, 34, 11, 40, 17, 36, 14, 8, 29, 30, 26, 23, 6	32, 9, 4, 25	2	24, 12, 16, 31, 37, 18, 39, 15, 3	0	400	mV	16
R2	"	5, 34, 11, 40, 17, 36, 14, 8, 29, 30, 26, 23, 9	32, 6, 4, 25	22	"	"	"	"	"
R3	"	5, 34, 11, 40, 17, 36, 14, 8, 29, 30, 26, 23, 4	32, 6, 9, 25	21	"	"	"	"	"
R4	"	5, 34, 11, 40, 17, 36, 14, 8, 29, 30, 26, 23, 25	32, 6, 9, 4	27	"	"	"	"	"
R5	Output crosstalk, low level	5, 34, 11, 40, 17, 36, 14, 8, 29, 30, 26, 23, 25	32, 6, 9, 4	28	"	"	600	"	"
R6	Output crosstalk, high level	5, 34, 11, 40, 17, 36, 14, 8, 29, 30, 26, 15, 31	3, 37, 18, 39	13	24, 12, 16, 25, 4, 9, 6, 23, 32	"	400	"	"
R7	"	5, 34, 11, 40, 17, 36, 14, 8, 29, 30, 26, 15, 37	3, 31, 18, 39	35	"	"	"	"	"
R8	"	5, 34, 11, 40, 17, 36, 14, 8, 29, 30, 26, 15, 18	3, 31, 37, 39	33	"	"	"	"	"

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* TABLE XIX. Output crosstalk (high-level and low-level) test - Continued.

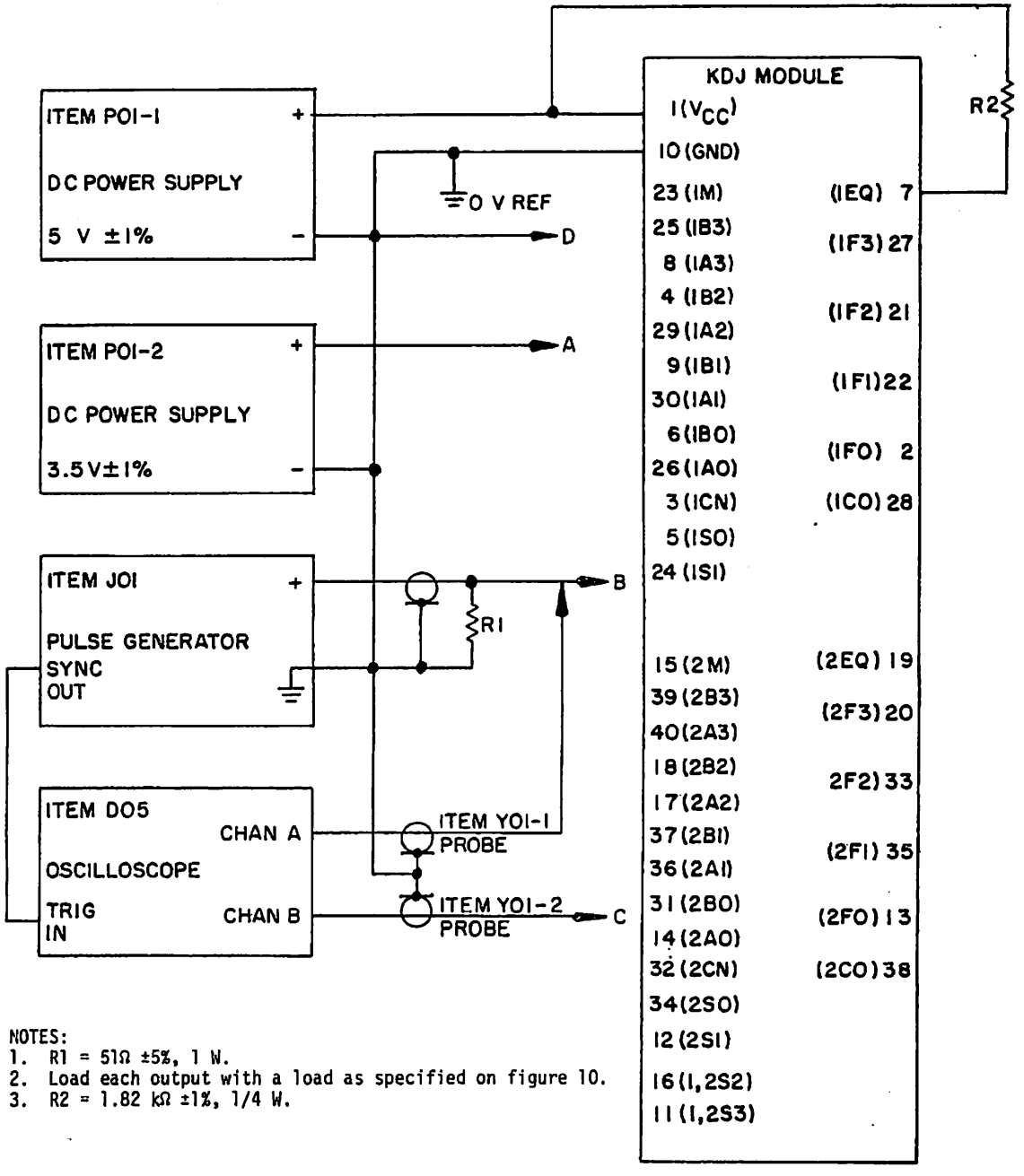
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R9	Output crosstalk, high level	5, 34, 11, 40, 17, 36, 14, 8, 29, 30, 26, 15, 39	3, 31, 37, 18	20	24, 12, 16, 25, 4, 9, 6, 23, 32	0	400	mV	16
R10	Output crosstalk, low level	5, 34, 11, 40, 17, 36, 14, 8, 29, 30, 26, 15, 39	3, 31, 37, 18	38	24, 12, 16, 25, 4, 9, 6, 23, 32	"	600	"	"
R11	"	5, 34, 11, 40, 17, 36, 14, 3	32, 30, 29, 8	2	24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6, 15, 23, 26	"	400	"	"
R12	"	"	32, 26, 29, 8	22	24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6, 15, 23, 30	"	"	"	"
R13	"	"	32, 26, 30, 8	21	24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6, 15, 23, 29	"	"	"	"
R14	"	"	32, 26, 30, 29	27	24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6, 15, 23, 8	"	"	"	"
R15	Output crosstalk, high level	"	"	28	24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6, 15, 23, 8	"	"	"	"
R16	Output crosstalk, low level	5, 34, 11, 36, 40, 17, 14, 3	"	7	24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6, 15, 23, 8	"	"	"	"

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* TABLE XIX. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R17	Output crosstalk, low level	5, 34, 11, 8, 29, 30, 26, 32	3, 36, 17, 40	13	24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6, 15, 23, 14	0	400	mV	16
R18	"	"	3, 14, 17, 40	35	24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6, 15, 23, 36	"	"	"	"
R19	"	"	3, 14, 36, 40	33	24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6, 15, 23, 17	"	"	"	"
R20	"	"	3, 14, 36, 17	20	24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6, 15, 23, 40	"	"	"	"
R21	"	"	3, 14, 36, 17	19	24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6, 15, 23, 40	"	"	"	"
R22	Output crosstalk, high level	"	3, 14, 36, 17	38	24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6, 15, 23, 40	"	"	"	"
R23	"	5, 34, 11, 8, 3, 29, 30, 26, 14, 36, 17, 40	32	7	15, 23, 24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6	"	"	"	"
R24	"	5, 34, 11, 8, 29, 30, 32, 26, 14, 36, 17, 40	3	19	15, 23, 24, 12, 16, 31, 37, 18, 39, 25, 4, 9, 6	"	"	"	"

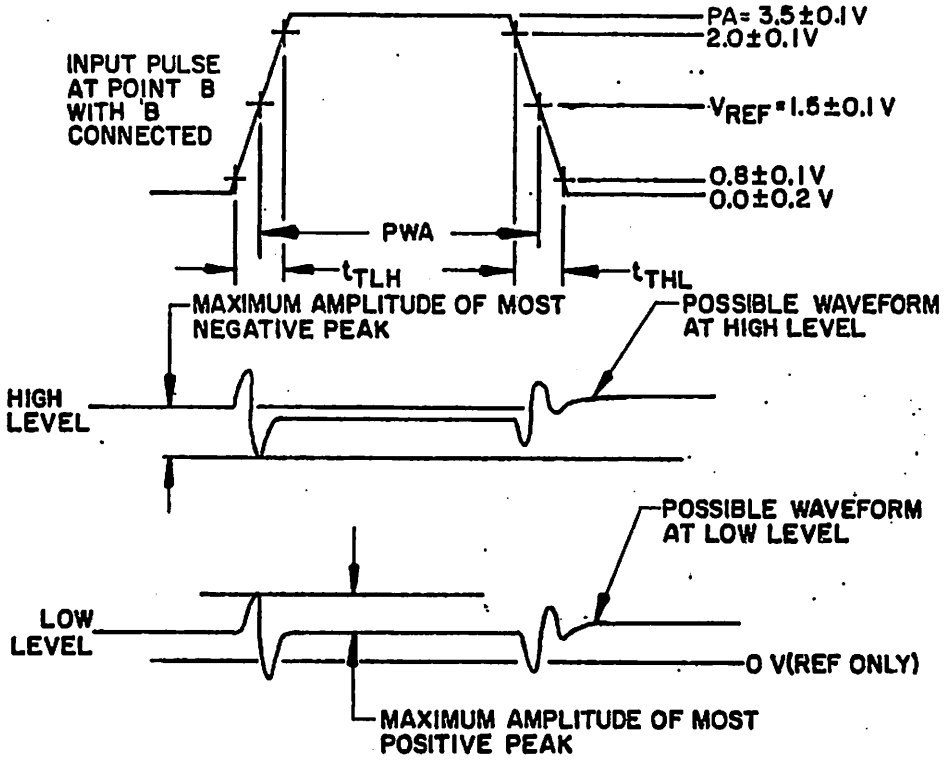
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load each output with a load as specified on figure 10.
3. R2 = 1.82 kΩ ±1%, 1/4 W.

* FIGURE 16. Output crosstalk (high-level and low-level) test circuit and waveforms for table XIX.

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NOTE: Input pulse parameters:

a. PWA = 500 ±50 ns.
b. $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
c. PRR = 1 MHz ±10%.

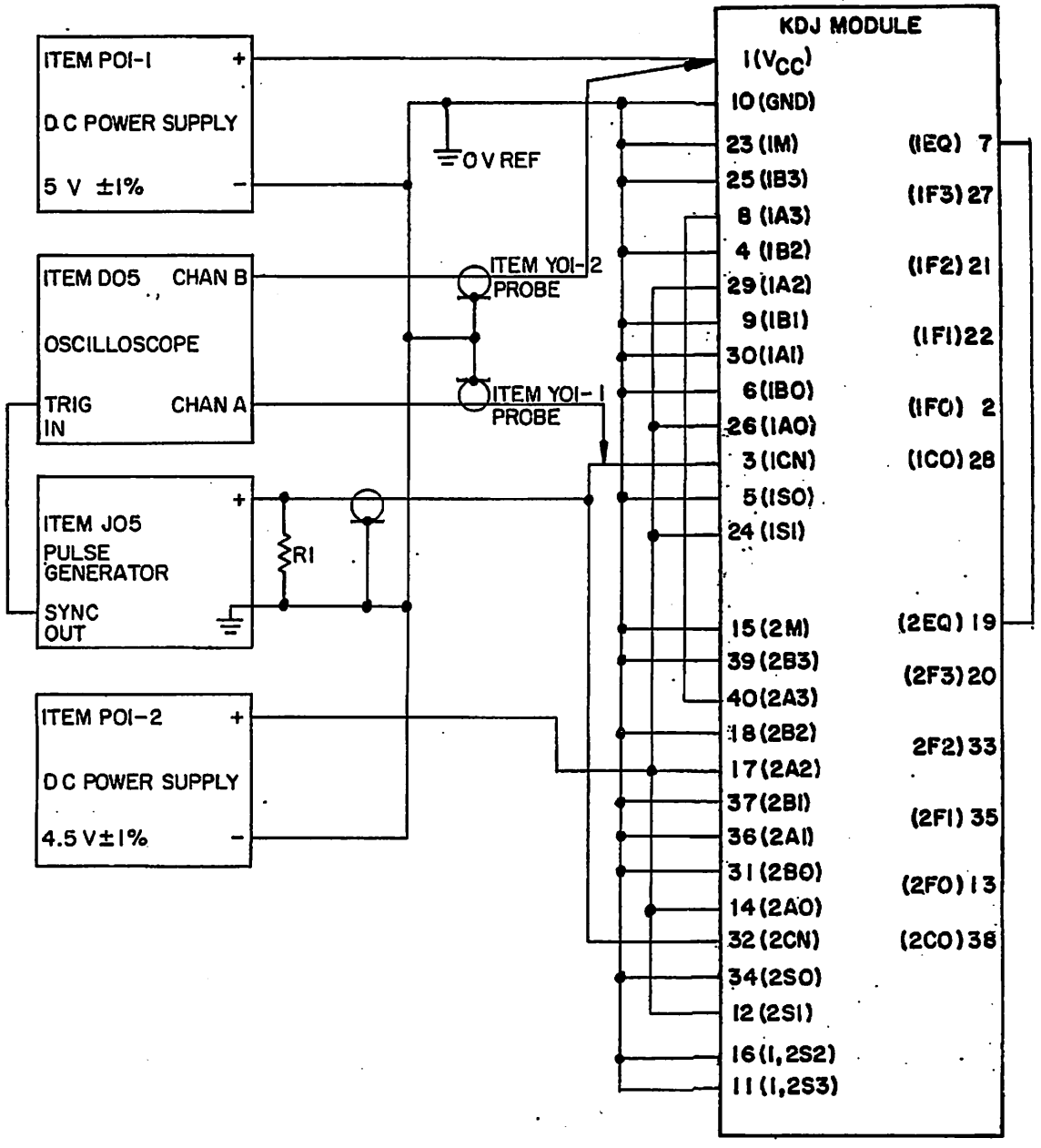
* FIGURE 16. Output crosstalk (high-level and low-level) test circuit and waveforms for table XIX - Continued.

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TABLE XX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	400	mV	17

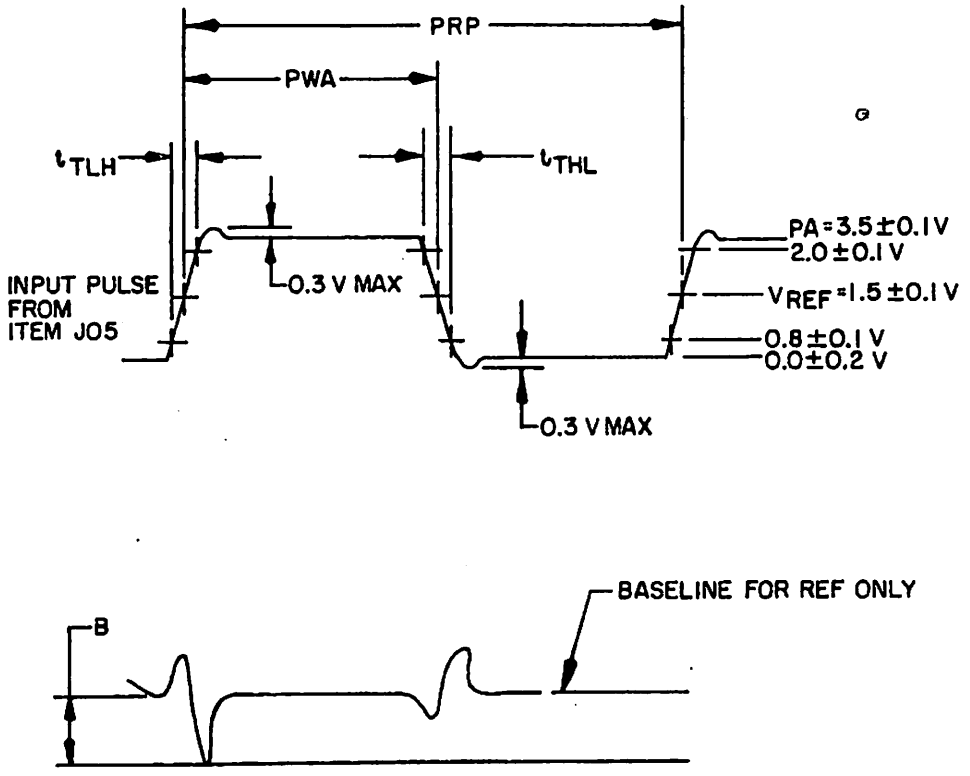
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 17. Power supply transient amplitude test circuit and waveforms for table XX.

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NOTES:
1. Input pulse parameters (item J05):
   PWA = 500 ±50 ns.
   PRP = 1000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
2. B = measurements from the baseline to the most negative excurson of the waveform.

FIGURE 17. Power supply transient amplitude test circuit and waveforms for table XX - Continued.

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6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - EC

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